Entrepreneurs and Housing Boom-Bust Dynamics*

Juhana Siljander†

Imperial College London‡

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Abstract

This paper builds a dynamic general equilibrium housing model to propose a new mechanism for explaining how credit supply shocks can generate housing boom-bust episodes. The model highlights how investors' capital allocation links to real estate dynamics, focusing on credit-constrained firms rather than credit-driven household demand. The model embeds a credit cycle, which generates an endogenous bust without an explicit shock reversal. Additionally, the model explains several other typical housing macro dynamics, such as house price momentum, large consumption elasticities out of housing wealth, and decreasing interest rates during a sustained house price boom. Quantitatively, the model can match the observed U.S. house price increase during the early 2000s.

JEL codes: R30, E30, E44, O47

Keywords: Credit Shock, House Prices, House Price Volatility, Housing Boom, Portfolio Allocation, Collateral Channel, Entrepreneurship

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‡Imperial College Business School Department of Finance. Contact: j.siljander@imperial.ac.uk

1 Introduction

Real estate bubbles and subsequent market crashes are often preceded by financial liberalization. For example, the housing booms and banking crises in Finland, Norway, and Sweden in the late 1980s and early 1990s, the "Asian financial crisis" in the mid-1990s, the "Japanese asset price bubble" of 1986 - 1991, and the housing boom of early 2000s in the U.S.A. and much of Europe. Discussing asset price bubbles, Kindleberger and Aliber (2011) write that while "most expansions of money and credit do not lead to mania … every mania has been associated with the expansion of credit."

However, the extent to which credit supply shocks can drive quantitatively sizable housing booms in economic models is still debated. In the context of the U.S. Great Recession, a series of recent papers have developed mechanisms where credit relaxation leads to a housing boom (Favilukis, Ludvigson, and Van Nieuwerburgh, 2017; Justiniano, Primiceri, and Tambalotti, 2019; Greenwald, 2018).[1] Another line of work criticizes these models for unrealistic assumptions, such as a lack of rental markets, and argues that (irrational) beliefs drove house prices (Iacoviello and Neri, 2010; Kiyotaki, Michaelides, and Nikolov, 2011; Justiniano, Primiceri, and Tambalotti, 2015; Kaplan, Mitman, and Violante, 2020). The question remains: Why do house prices fluctuate substantially in the data when generating such fluctuations is challenging in economic models? Piazzesi and Schneider (2016) call this the "house price volatility puzzle."

I tackle these questions by proposing a new theoretical mechanism for generating significant house price effects from credit shocks. Contrary to much of the existing quantitative theory literature, which has concentrated on household demand (Mian et al., 2013; Mian and Sufi, 2014; Mian et al., 2017), I emphasize the role of credit-constrained entrepreneurs, relating to another strand of the empirical literature considering how financial frictions affect capital allocation (Whited, 2006; Buera et al., 2011; Chaney et al., 2012; Matvos and Seru, 2014; Robb and Robinson, 2014; Cvijanović, 2014; Adelino et al., 2015, 2016; Liebersohn, 2017; Schmalz et al., 2017; Bahaj et al., 2020; Whited and Zhao, 2021). The mechanism builds on two observations highlighting the special nature of housing investments.

[1] U.S. household debt approximately doubled during 2000 – 2007, largely fueled by an increase in private-label securitization of mortgage backed securities (MBS), which contributed 46% of the MBS market in dollar volume in 2004, up from 22% in 2003 (Levitin and Wachter, 2011). Loan-to-value (LTV) ratios for new mortgage originations also rose from approximately 85 % in 2000 to 95% in 2005 (Duca, Muellbauer, and Murphy, 2011).

First, similarly to many financial assets such as bonds, the expected return on housing is mostly homogeneous across different investors. This feature is unlike many other types of business capital, whose productivity can vary substantially across entrepreneurs (Fagereng et al., 2020).[2] Consequently, when productive entrepreneurs expand their businesses, investors disadvantaged in operating business capital accommodate the entrepreneurs' capital demand by rebalancing their portfolios towards housing (and other financial assets), where they are not similarly disadvantaged. Therefore, the price impacts of such a capital demand shock and the subsequent investor rebalancing depend on who the marginal investors are, whose portfolio choices drive asset prices, and how elastic their supply of capital is to more productive investors. In particular, a demand shock for productive capital affecting capital prices can have an equally substantial house price effect as the demand shock propagates to the housing market via this *portfolio rebalancing channel*.

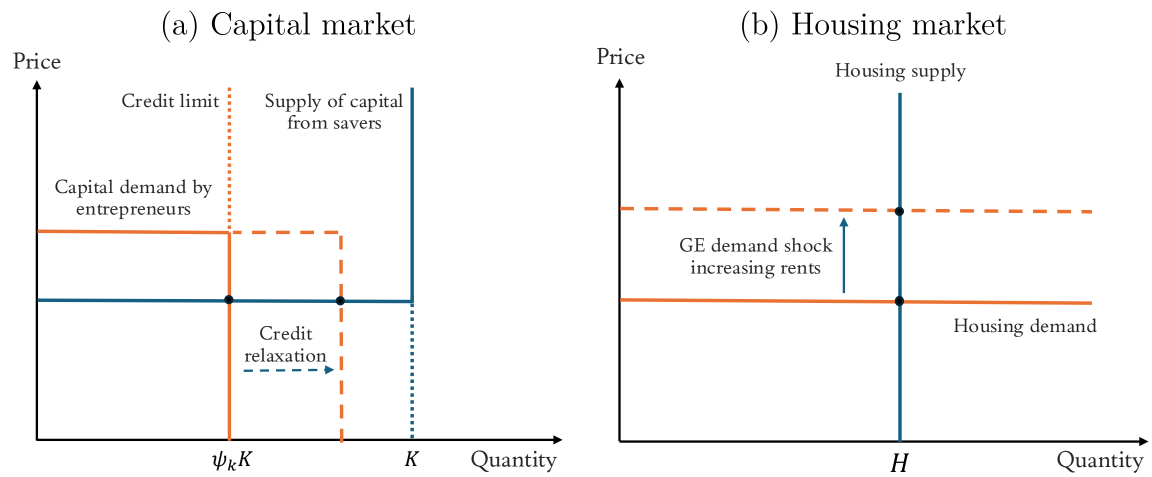
Second, the housing rental price depends on the productivity of other assets via the *general equilibrium demand channel*. In a heterogeneous agent setting, the collateral credit constraint affects capital allocation and aggregate productivity (Buera et al., 2011; Buera and Moll, 2015; Buera and Nicolini, 2020). Increased income increases demand for all kinds of consumption, including housing services. With a fixed housing supply (at least in the short run), rents rise, increasing house prices. Therefore, the critical feature underlying my mechanism is that credit markets can affect productivity by changing the allocation of factor inputs, as productive but credit-constrained entrepreneurs get access to loans (see Restuccia and Rogerson (2017) and the references therein).

I illustrate this house price mechanism in Figure 1, which depicts the propagation of the credit shock from the capital market to the housing market via the general equilibrium demand effect. The figure relates to a static simplification of my model with two types of agents—productive *entrepreneurs* and less productive *savers*—presented in detail in Appendix A. Due to their higher productivity, entrepreneurs have a higher valuation for business capital than savers. The credit constraint, however, limits their capital holdings. A credit relaxation allows them to purchase business capital, which the savers supply to them perfectly elastically. The perfectly elastic capital supply by

[2] Holding period returns on housing investments are known to vary substantially *ex post*, largely driven by the substantial heterogeneity in purchase prices relative to the "fair" value of the property. However, the high variance in ex post returns is different from what an investor can expect *ex ante*. In contrast, Fagereng et al. (2020, Table 7) report that returns to net worth among business owners are highly persistent, and they exhibit substantial heterogeneity with a three-fold difference between the median and the 90th percentile.

Figure 1: House price amplification for a credit shock

The figure demonstrates the propagation of the credit shock from the capital market to the housing market: (a) with linear $\underline{a}k$ technology, the savers' valuation for the capital is given by the productivity parameter $\underline{a}$. When faced with a demand shock for capital by the entrepreneurs, they provide perfectly elastic supply of capital to entrepreneurs up until they have exhausted their capital holdings, and the entrepreneurs own all of the capital stock. The perfectly elastic supply prohibits any price effects on capital; (b) the savers' valuation for housing, however, increases with higher rents, increasing house prices.



savers prevents any price effects on business capital. In contrast, given the improved capital allocation associated with higher aggregate output, house prices rise due to the general equilibrium demand effect, which increases rents.[3]

My primary analysis in section 2 concerns a dynamic model, where credit relaxation lifts the entrepreneurs completely off their credit constraints (cf. occasionally binding constraints). Like the static setup, house prices increase, but with an additional amplification increasing price-to-rent ratios: the disproportionate price appreciation compensates investors for the lower rental yields. Moreover, the credit relaxation leads to a sustained boom, where house prices keep growing even after the original shock impact.

Quantitatively, calibrating the housing supply elasticity with recent investment and rent data and the credit shock with observed changes in loan-to-value ratios for new loan originations produces a 52.9% increase in house prices, thus fully explaining the 40.0% price increase during the "house price bubble" in Q1/2003 – Q1/2007. This suggests that this house price mechanism emphasizing agents' portfolio incentives is quantitatively powerful, even if it was unlikely to be the only contributor to the house

[3] I relax the assumption on perfectly elastic capital supply for the static model in Appendix A.4 and for the dynamic model in section 2.

price boom. In particular, the fact that the model can produce a disproportionate increase in house prices relative to other asset prices suggests it may provide a potential solution to the "house price volatility puzzle" of Piazzesi and Schneider (2016).

However, while quantitatively substantial, the house price boom is nevertheless short-lived. The entrepreneurs' higher consumption propensity decreases their relative wealth during the transition following the shock. Eventually, this leads to a reversal, where the entrepreneurs become credit-constrained and liquidate their capital holdings to finance higher consumption.

Once the entrepreneurs hit the credit constraint, the interest rate drops discontinuously, and output decreases due to deteriorating capital allocation. In a frictionless setup, these "flight-to-safety" dynamics—resembling the market crash during the U.S. financial crisis—violate typical asset pricing equations: the stochastic discount factor and interest rate cannot decrease simultaneously in a frictionless setup. Guerrieri and Lorenzoni (2017) solve this problem by considering an exogenous credit tightening, which generates a wedge into the asset pricing equation. Mian and Sufi (2014) have, however, argued that the financial crisis was not associated with credit tightening. In my model, the bust is an endogenous outcome of the original credit shock.

Since my mechanism can generate an entire boom-bust episode with a single, one-time shock in a rational expectations model and without an explicit shock reversal, I suggest it is a valuable complement to existing explanations of the boom-bust episode.

Notably, my mechanism is robust to including frictionless rental markets and housing investments. The key difference relative to previous work is that I emphasize the role of entrepreneurial credit constraints on business capital instead of the credit constraint on housing. In fact, in my model, relaxation of the housing collateral constraint does not affect asset prices, replicating one of the leading quantitative/empirical findings of Kaplan, Mitman, and Violante (2020), who argue that the share of financially constrained households is too low to generate a significant demand boom after a credit relaxation. Consistent with this argument, homeowners are not credit-constrained in equilibrium, rendering the housing collateral constraint irrelevant. Instead, the relevant collateral constraint concerns entrepreneurs' business capital. This setup provides a clean and transparent demonstration of why the entrepreneurial finance setting can generate a substantial house price effect for a credit shock when previous models concentrating on the household demand channel and housing constraints have produced the opposite result.

I also use my model to explain three additional findings in the existing literature. First, the model produces large consumption elasticities out of housing wealth, similar to Berger, Guerrieri, Lorenzoni, and Vavra (2017); I replicate their main result (Berger et al., 2017, Proposition 1), but now in the context of endogenous house prices, whereas they consider exogenous price shocks. Producing a house price boom to realistically calibrated exogenous shocks has been difficult in the macro housing literature, and the secular long-run increase in house prices has often been attributed to increased housing preferences, e.g., due to improved location-specific amenities. Shifting preferences would, however, shut down the substitution channel crucial for producing the large consumption elasticities for house price shocks; I contribute to this literature by endogenizing the price boom while still maintaining the large consumption elasticities for house price shocks.

Second, as my model produces a sustained boom after a credit shock, it generates house price momentum similar to Guren (2018) and consistent with the highly positive autocorrelation observed in the house price time series data (Case and Shiller, 1988). In particular, my mechanism generates house price momentum in a rational expectation setup, where all agents expect high house price growth in the future but do not act on the information in a manner that would counteract the price forces. This occurs because the marginal agents determining house prices are fully invested in the housing market, whereas the other agents have even better investment opportunities that are exclusively accessible to them. My model, therefore, highlights the special nature of heterogeneous agent models, where different investors' varying portfolio incentives can generate highly non-trivial asset price dynamics.

Third, while the unanticipated credit relaxation generates a positive interest rate shock at impact, it is followed by a long period of decreasing interest rates during the transition path, consistent with the U.S. data during the housing boom of the early 2000s. This occurs due to the different consumption propensities between the agents. The agents with low consumption propensities and, therefore, high savings rates accumulate wealth, implying that the wealth-weighted average consumption propensity decreases. I use closed-form characterizations to demonstrate that this consumption propensity determines the risk-free interest rate during the transition, resembling the main result of Mian, Straub, and Sufi (2021): decreasing aggregate consumption propensity is associated with increasing aggregate savings rate, which drives up bond prices and lowers the interest rate.

I conclude the paper with a brief empirical evaluation of the macroeconomic significance of the mechanism, augmenting my quantitative model analysis. In Figure 2, I use Community Reinvestment Act (C.R.A.) bank-county-level loan data to demonstrate strong correlations between collateral valuations, business loan originations,

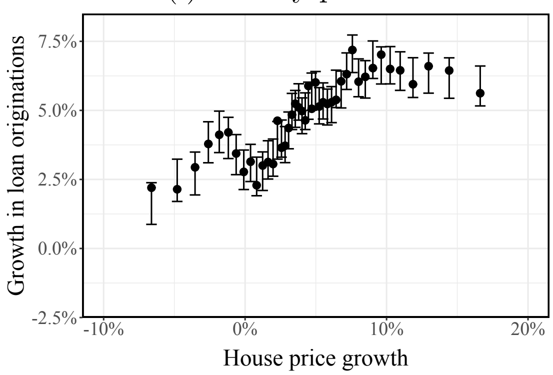
Figure 2: Collateral channel, business loans, and business activity

Panel A plots Cattaneo et al. (2024) binscatter between house prices and business loan originations (≤ \$250,000 at origination). The left-hand side graph uses county-level data, whereas the right-hand side figure uses bank-county-level data during 2000 – 2022. House price data is from Zillow, and business loan originations are from Community Reinvestment Act data provided by the U.S. Federal Financial Institutions Examination Council. Panel B shows similar binscatters documenting the relation between small business loan originations and county-level annual growth rates in the number of establishments and employment in County Business Patterns data. The establishment and employment counts exclude residential construction and real estate sectors.

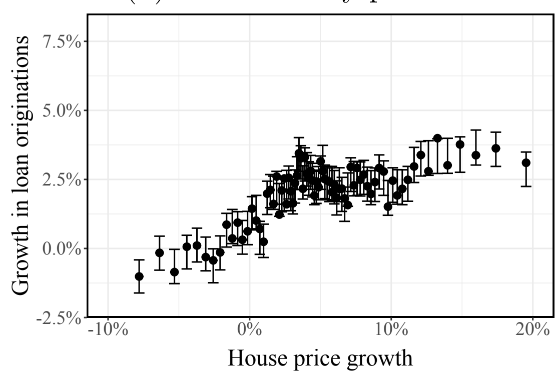
Panel A

House prices and small business loan origination

(i) County panel



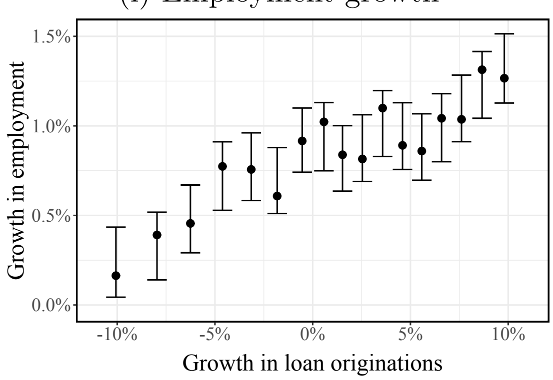
(ii) Bank-County panel



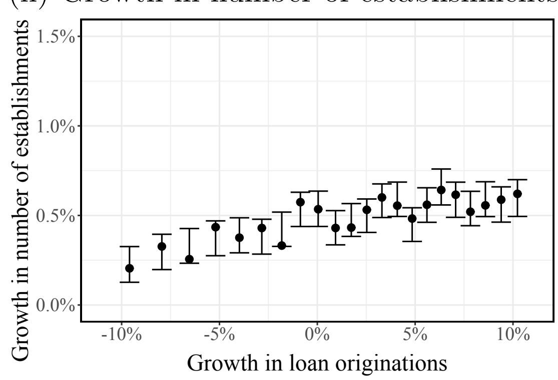
Panel B

Small business loans origination and business activity

(i) Employment growth



(ii) Growth in number of establishments



and business activity (employment, number of establishments). In the empirical Appendix B, I further document that these correlations are robust to instrumenting house price or loan growth rates with the Saiz (2010) land topography instrument. The estimates I obtain are broadly consistent with those of Adelino, Schoar, and Severino (2015), who argue that the increase in house prices explained 10 – 25% of the aggregate U.S. employment growth during the pre-crisis years via the "collateral channel." By leveraging the C.R.A. data, my empirical contribution in Appendix B completes the chain from increased collateral valuations to *increased loan originations* to increased business activity, consistent with the widely understood role of collateral/land valuations for business financing and the aggregate economy, as discussed, e.g., by Liu, Wang, and Zha (2013) and Adelino et al. (2015).[4]

Finally, I finish the feedback loop by considering the effect of employment on house prices. Existing literature suggests a large income elasticity of housing demand (Davis and Ortalo-Magné, 2011), implying that increases in aggregate productivity should generate substantial rent effects. In section 6, I instrument county-level employment growth rates with the Bartik instrument and combine the obtained elasticity with the above estimates on how collateral valuations affect employment. Together, these provide a simple back-of-the-envelope evaluation of the total feedback mechanism. Combined with the quantitative analysis of my model, these analyses suggest that my mechanism may help understand how credit conditions may generate substantial house price effects.

My paper is related to the empirical (Mian and Sufi, 2009; Levitin and Wachter, 2011; Mian et al., 2020) and quantitative theory literature (Iacoviello, 2005; Kiyotaki, Michaelides, and Nikolov, 2011; Justiniano, Primiceri, and Tambalotti, 2015, 2019; Favilukis, Ludvigson, and Van Nieuwerburgh, 2017; Greenwald, 2018; Greenwald and Guren, 2021) on how credit conditions affect house prices. I propose a new price amplification mechanism, where a credit shock propagates through the financial markets to generate a housing boom. Concerning such mechanisms, the Kiyotaki and Moore (1997) financial accelerator has been previously considered in models with housing or land by Iacoviello (2005) and Liu, Wang, and Zha (2013). I augment this financial accelerator literature by documenting a novel amplification mechanism building on a

[4] Adelino et al. (2015) study commercial and industrial (C&I) lending in bank Call Reports and do not find a statistically significant association between house prices and C&I lending. In Appendix B, I augment this analysis by using the C.R.A. business loan data to document a statistically and economically significant effect of house price growth also on business lending, in addition to the effects on home equity refinancing and cash-out home equity lines of credit (HELOCs) documented by Adelino et al. (2015).

specific interaction between the capital and housing markets.[5]

The mechanism builds on two key observations: (i) financial constraints affect capital allocation and productivity, an idea discussed earlier in the theoretical (Buera et al., 2011; Moll, 2014; Buera and Moll, 2015; Buera and Nicolini, 2020) and empirical literature (Whited, 2006; Gomme et al., 2011; Buera et al., 2011; Chaney et al., 2012; Matvos and Seru, 2014; Robb and Robinson, 2014; Cvijanović, 2014; Adelino et al., 2015, 2016; Liebersohn, 2017; Schmalz et al., 2017; Bahaj et al., 2020; Doerr, 2020, 2021; Whited and Zhao, 2021), and (ii) shocks to the constraint propagate through financial markets as investors rebalance their portfolios. The latter relates to literature considering investors' and households' portfolio choices during housing boom-bust episodes (Pennacchi and Rastad, 2011; Martinez-Toledano, 2020; DeFusco et al., 2022; Gargano and Giacoletti, 2022).

The paper is organized as follows. Section 2 presents the dynamic housing model. Section 3 includes the main theorems and pricing decompositions, incl. and additional discussion of the empirical relevance of my model in section 3.1. Section 4 discusses the credit shock mechanism in general equilibrium. Section 5 produces the quantitative analysis with an estimate of the housing supply elasticity used in model calibration. Section 6 discusses testable predictions of my mechanism in reduced form regressions. Section 7 concludes.

2 Model

I present a dynamic general equilibrium model consisting of capital, housing, rental, bond, and goods markets. The model introduces housing markets into an entrepreneurial finance model (Evans and Jovanovic, 1989) with an heterogeneous agents setup, where credit conditions affect capital allocation and aggregate productivity (Buera and Moll, 2015). The model includes two types of agents: productive *entrepreneurs* and less productive *savers*. The heteregeneity in the agents' productivity implies that capital allocation matters for aggregate output and housing rents. The house price effect of capital allocation depends on who is the marginal agent pricing the assets at any particular time. The dynamic reallocation of capital may affect the identity of the marginal agent, generating highly non-linear price effects both at impact and during the transition, after an exogenous unanticipated credit shock affecting capital allocation.

[5] In Appendix A.4.2, I use a simplified version of the model to illustrate a novel strategic interaction between the housing and capital markets, embedded in my model mechanism.

The main experiment that I consider is an unanticipated exogenous relaxation of loan-to-value (LTV) collateral credit constraint affecting the entrepreneurs ability to finance their business activities. The credit relaxation triggers capital reallocation as the productive entrepreneurs purchase capital from the savers. Aggregate productivity increases due to the entrepreneurs higher productivity. This drives up rents and house prices.

On the other hand, the direct shock on capital demand—induced by the credit relaxation—is partially absorbed by the savers, who supply capital to the entrepreneurs, and rebalance their portfolios toward housing. The elasticity of this capital supply by savers to entrepreneurs determine the capital price effect. In contrast, the house price effect depends on the credit shocks' effect on rents and on the expected future price appreciation rates of capital and housing. I show that these forces can generate substantial house price amplification, generating a disproportionate increase in house prices.

In a simplified static set-up, I document these mechanics in more detail in Appendix A.4 using closed-form solutions, whereas my main analysis in this section concerns a fully dynamic setup. At the high level, my mechanism documents that the general equilibrium demand channel, where increased productivity affects house prices via increased rents, can generate substantial price effects. This relates to the well-known fact that in heterogeneous agent models, the indirect, general equilibrium responses may dominate the direct effects (Kaplan, Moll, and Violante, 2018).

2.1 Environment

Time is continuous and horizon is infinite. Two types of agents populate the economy: savers indexed by $j \in [0,1]$ and entrepreneurs indexed by $j \in (1,2]$. All agents maximize lifetime utility

$$\int_o^\infty e^{-\rho^j t} u(c_t^j, s_t^j)\, dt,$$

where c_t^j and s_t^j, respectively, denote non-durable consumption and housing services. Each agent produces the consumption good with a linear $a^j k_t$ production technology.[6] Housing services s_t are produced using housing capital h_t and are traded freely on a rental market at a rental price r_{st}. Housing and productive capital are in fixed supply

[6] In Appendix A.4, I relax this assumption, in a simplified static version of the model, by allowing decreasing returns to scale with the production function specified as $ak^\gamma, \gamma < 1$. I document that the main price amplification mechanism discussed in this paper is robust for such a generalization.

H and K, respectively, and are both homogeneous.[7]

The agents can freely trade housing, productive capital as well as a risk-free bond b_t, which pays interest r_{ft}. However, they do face a short-selling constraints $k_t^j \geq 0$ and $h_t^j \geq 0$ as well as a collateral constraint given by

$$-b_t^j \leq (1-\phi_k)p_{kt}k_t^j + (1-\phi_h)p_{ht}h_t^j, \quad \phi_k, \phi_h \in [0,1], \tag{2.1}$$

where p_{kt} and p_{ht} denote the prices of capital and housing, respectively.[8]

The following two assumptions specify the type of heterogeneity I consider.

Assumption 2.1. Savers discount future at rate $\rho^j = \underline{\rho}$ for all $j \in [0,1]$, and entrepreneurs discount future at rate $\rho^j = \rho > \underline{\rho}$ for all $j \in (1,2]$.

Assumption 2.1 says that savers are more patient than entrepreneurs. Importantly, this implies that savers have larger saving rates. I abstract away from intermediaries and the banking sector, in general, and in equilibrium savers will provide the financing for entrepreneurs. The savers correspond to bondholders in the data: most of the wealth used to finance entrepreneurial activity is held by individuals in the top decile of the wealth distribution and insurance and pension funds.

Assumption 2.2. Savers have productivity $a^j = \underline{a}$ for all $j \in [0,1]$, and entrepreneurs have productivity $a^j = a > \underline{a}$ for all $j \in (1,2]$.

Assumption 2.2 says that entrepreneurs are more productive than savers in producing the consumption goods. One interpretation is that the entrepreneurs are a set of wealthy hand-to-mouth individuals, who have good investment opportunities, but limited ability to finance all of them.[9] More generally, however, the crucial feature this assumption delivers is that the credit constraint affects capital allocation and aggregate output; while I discuss this in the context of entrepreneurial financing, alternative interpretations delivering this main feature would suffice.

[7]The quantitative results are robust for also including housing investments, but for simplicity I only include them in my quantitative analysis in Section 5.

[8]Note that all debt is short-term; I concentrate on business loans, which typically have variable interest rates and many such loans have covenants allowing the lender to force loan pre-payment if a collateral condition is violated. For instance, Glennon and Nigro (2005) document that in their sample of the Small Business Administration (SBA) 7(a) loan guarantee program 84% of loans had variable interest rate. Moreover, Chodorow-Reich and Falato (2017) report that 37% of loans in the Shared National Credit Program data were terminated or reduced before maturity during the crisis years of 2008 − 2009, approximately matching the fraction of loans with a covenant violation, consistent with forced loan prepayment. See also Falato and Liang (2016).

[9]An important implicit friction in this set-up is that savers cannot hire the entrepreneurs to operate their capital, but the actual ownership structure matters (Hart and Moore, 1994).

2.2 Optimization problem

I denote an agent's net worth by $n_t^j = p_{kt}k_t^j + p_{ht}h_t^j + b_t^j$, where the agent chooses her portfolio (k_t^j, h_t^j, b_t^j).

I define the return on capital and housing, respectively, as

$$r_{kt}^j \equiv \underbrace{\frac{a^j}{p_{kt}}}_{\text{Dividend yield}} + \underbrace{\frac{dp_{kt}}{p_{kt}}}_{\substack{\text{Capital price}\\ \text{appreciation}}} \quad \text{and} \quad r_{ht} \equiv \underbrace{\frac{r_{st}}{p_{ht}}}_{\text{Rental yield}} + \underbrace{\frac{dp_{ht}}{p_{ht}}}_{\substack{\text{House price}\\ \text{appreciation}}}.$$

Note that the key difference between capital and housing assets in the model is the heterogeneity in capital returns (Fagereng, Guiso, Malacrino, and Pistaferri, 2020).

In what follows, I will use the following notations for the capital and house price appreciation rates, respectively:

$$\frac{dp_{kt}}{p_{kt}} \equiv \mu_{kt}^p \, dt \quad \text{and} \quad \frac{dp_{ht}}{p_{ht}} \equiv \mu_{ht}^p \, dt$$

For simplicity, I will now drop the superscripts j, unless required to highlight the heterogeneity in the model. I denote the portfolio shares of capital and housing, respectively, by

$$\theta_{kt} = \frac{p_{kt}k_t}{n_t} \quad \text{and} \quad \theta_{ht} = \frac{p_{ht}h_t}{n_t}.$$

With this notation I rewrite the collateral constraint (2.1) as

$$\phi_k\theta_{kt} + \phi_h\theta_{ht} \leq 1. \tag{2.2}$$

Using this notation, an agent's wealth evolves as

$$\frac{dn_t}{n_t} = \underbrace{-\frac{c_t}{n_t} - \frac{r_{st}s_t}{n_t}}_{\text{Consumption}} + r_{ft} + \theta_{kt}\underbrace{\left[r_{kt}^j - r_{ft}\right]}_{\substack{\text{Excess return}\\ \text{on capital}}} + \theta_{ht}\underbrace{\left[r_{ht} - r_{ft}\right]}_{\substack{\text{Excess return}\\ \text{on housing}}} dt. \tag{2.3}$$

The first two terms on the wealth evolution describe consumption of consumption goods and housing services. On the other hand, the agent's portfolio return is given by the risk-free rate and the excess returns on capital and housing investments.

I look for a recursive competitive equilibrium driven by an aggregate state vector $\mathbf{S}$. Observe that given the aggregate state, an agent's opportunity set is fully determined by her net worth n, which is the agent's state. Given capital and house prices $p_{kt} = p_k(\mathbf{S}_t), p_{ht} = p_h(\mathbf{S}_t)$, respectively, risk-free interest rate $r_{ft} = r_f(\mathbf{S}_t)$, rental price $r_{st} = r_s(\mathbf{S}_t)$ and a law of motion $d\mathbf{S}_t$ for the aggregate state $\mathbf{S}_t$, agents' policies for non-durable consumption c_t, housing consumption s_t as well as portfolio shares on capital θ_{kt} and housing θ_{ht} solve the agent's problem; she maximizes her lifetime utility:

$$V(n; \mathbf{S}) = \max_{\{c_t, s_t, \theta_{kt}, \theta_{ht}\}_{t \geq \tau}} \mathbb{E}_\tau \left[\int_\tau^\infty e^{-\rho t} u(c_t, s_t)\, dt \right]$$

where the maximization is subject to the wealth evolution (3.3), the LTV constraint (2.2) and short-selling constraints $\theta_{kt} \geq 0$ and $\theta_{ht} \geq 0$, given initial states $n_\tau = n$ and $\mathbf{S}_\tau = \mathbf{S}$.

2.3 Equilibrium

With homothetic preferences the individual savers' and entrepreneurs' problems aggregate, respectively, to those of two representative agents, one for each type. Correspondingly, I denote the type-specific aggregate quantities for savers and entrepreneurs, respectively, by

$$\underline{n}_t = \int_0^1 n_t^j\, dj \quad \text{and} \quad n_t = \int_1^2 n_t^j\, dj,$$

and similarly for $\underline{k}_t, k_t, \underline{h}_t, h_t, \underline{b}_t, b_t, \underline{c}_t, c_t$ and $\underline{s}_t, s_t$.

Definition 2.3 (Recursive competitive equilibrium)**.** A recursive competitive equilibrium consists of

(i) value functions $V^i(n^i; \mathbf{S})$;

(ii) policy functions $c^i, s^i, \theta_h^i, \theta_k^i$ depending on n^i and $\mathbf{S}$;

(iii) prices $p_h(\mathbf{S}), p_k(\mathbf{S}), r_s(\mathbf{S}), r_f(\mathbf{S})$, and

(iv) aggregate state evolution $d\mathbf{S} = \boldsymbol{\mu}^S dt$

such that

1. $V^i(n^i; \mathbf{S})$ and the above associated policy functions solve the individual HJB equations (C.1) given prices and state evolution.

2. Markets clear

   - Goods market: $c_t + \underline{c}_t = ak_t + \underline{ak}_t$;
   - Rental market: $s_t + \underline{s}_t = H$;

- Housing market: $h_t + \underline{h}_t = H$;
- Capital market: $k_t + \underline{k}_t = K$;
- Bond market: $b_t + \underline{b}_t = 0$.

3. The aggregate state evolution $d\mathbf{S}$ is consistent with agents' equilibrium choices, given policy functions and market clearing conditions.

I denote aggregate consumption by $C = c + \underline{c}$ and the total wealth in the economy by $N_t = \underline{n}_t + n_t = p_{kt}K + p_{ht}H$. Let

$$\eta_t = \frac{n_t}{\underline{n}_t + n_t} \quad \text{and} \quad \nu_t = \frac{p_{kt}K}{N_t}$$

denote, respectively, entrepreneurs' wealth share and the share of total wealth contributed to the productive capital. I look for a symmetric equilibrium, where entrepreneurs and savers have identical portfolio shares within a type, and with the aggregate state $\mathbf{S} = \eta$.

In the sequel, I denote the entrepreneurs' share of capital and housing ownerships, respectively, by

$$\psi_{kt} = \frac{k_t}{K} \quad \text{and} \quad \psi_{ht} = \frac{h_t}{H}.$$

3 Model characterization

I begin characterizing the model equilibrium with the following result, theoretically replicating one of the leading quantitative/empirical findings in Kaplan et al. (2020).

Proposition 3.1 (Irrelevance of housing collateral)**.** *A financially constrained agent will never own housing, i.e., $h_t^j = 0$ necessarily. Consequently, relaxation of the housing collateral constraint ϕ_h does not affect house prices.*

The above result is clearly unrealistic, and it would not be challenging to relax it by extending the model.[10] Nevertheless, this result speaks directly to the heart

[10]For example, one could add behavioral friction requiring each model agent to own some housing for their own consumption. As I will show, the model mechanism produces a disproportionate increase in house prices due to a credit shock. Therefore, such friction would merely amplify the model mechanism, as the entrepreneur would benefit from the disproportionate rise in the housing collateral value (Kiyotaki and Moore, 1997; Doerr, 2020). In a slightly different context, this statement is made precise by Theorem A.11 in Appendix A.4.2, where I show in a static set-up that for entrepreneurs whose initial portfolio consists of housing, the amplification effect is magnified, possibly even producing multiple equilibria due to a new strategic interaction between the capital and housing markets.

of the debate on the effects of credit conditions on house prices. Most existing work concentrates on the effects of housing collateral, where a relaxation of such a collateral constraint supposedly generates a housing boom via the demand channel. However, Kaplan et al. (2020) argue that, empirically, the set of financially constrained households is not large enough to produce a substantial housing boom via the demand channel. Proposition 3.1 replicates their finding in the extreme setup, where no constrained agent owns any housing. In this model, the collateral constraint in housing investments is entirely irrelevant, as no agents who own housing and might be affected by such a credit relaxation are financially constrained. Therefore, this proposition provides full confidence that none of the following analyses relies on the housing collateral, but all of the following theoretical analyses only concern entrepreneurial credit constraints. In particular, this result highlights that the mechanism I develop is entirely orthogonal to the existing theory debate on the effect of credit conditions on house prices and, therefore, provides a novel way to approach the issue.

Regarding the underlying mechanism, in this model, an entrepreneur can be financially constrained only if their marginal productivity for capital exceeds that of the saver. Since the return on housing investments is the same for both agents and by the saver's no-arbitrage condition, that return must equal their marginal productivity on capital, a financially constrained entrepreneur will always enjoy better returns by increasing their capital investments over any housing investments. Therefore, in equilibrium, they will always choose not to invest in housing if the constraint is binding.

In order to illustrate the main feedback mechanism providing the house price amplification as well as the eventual bust, I next provide closed form decompositions for capital prices.

Proposition 3.2. *Price of productive capital can be expressed as*

$$p_k = \frac{A(\psi_k)}{\frac{c}{n}\eta + \frac{c}{\underline{n}}(1-\eta)}\nu$$

and house price p_h as

$$p_h = \frac{A(\psi_k)}{\frac{c}{n}\eta + \frac{c}{\underline{n}}(1-\eta)}\frac{K}{H}(1-\nu),$$

where $A(\psi_k) = a\psi_k + (1-\psi_k)\underline{a}$ is the average productivity of capital.

Moreover, for utility of the form $u(c^\alpha s^{1-\alpha})$ satisfying Inada conditions, the housing rental rate is given by

$$r_s = A(\psi_k)\frac{K}{H}\frac{1-\alpha}{\alpha}.$$

In order to unpack the contents of this result, I begin by noting that the above theorem decomposes the rental rate of housing as follows:

$$r_s = \underbrace{A(\psi_k)}_{\substack{\text{Average productivity} \\ \text{of capital}}} \times \underbrace{\frac{K}{H}}_{\substack{\text{Supply of capital relative to} \\ \text{supply of housing}}} \times \underbrace{\frac{1-\alpha}{\alpha}}_{\substack{\text{Preferences for housing} \\ \text{vs. non-durable consumption}}} . \tag{3.1}$$

The average productivity $A(\psi_k)$, which depends on capital allocation ψ_k, and the supply of capital K, determines the economy's productive capacity. At the macro level, this determines aggregate income in the economy and, therefore, in equilibrium, determines the aggregate demand for all consumption, incl. housing services. The last term in the decomposition, $(1-\alpha)/\alpha$, describes how that demand is divided between housing demand and demand for non-durable consumption. Together, these two components thus describe the aggregate demand for housing services. Finally, this is contrasted with the housing supply, H, in the denominator of the second component. In all its simplicity, the rental rate is therefore given by the ratio of demand for and supply of housing services, measured in terms of the numéraire of the economy.[11]

Similarly, Theorem 3.2 also produces a closed-form solution for the price of housing. We observe two differences in the house price formula relative to the rental rate r_s. First, the denominator in the preference component is replaced by the aggregate consumption propensity

$$\frac{C}{N} = \eta\frac{c}{n} + (1-\eta)\frac{\underline{c}}{\underline{n}}. \tag{3.2}$$

The consumption propensity is one minus the savings rate; intuitively, a high savings rate means a high demand for assets, which drives up asset prices. Note that this object can, in principle, be observed in the data both at aggregate and micro levels. Furthermore, under some additional assumptions, in Lemma 4.2 I will characterize this using the model primitives.

We observe that fluctuations in the savings rate occur due to two reasons. On one hand, individual consumption propensities fluctuate together with the economy. On the other hand, wealth-weighted composition of agents with different saving propensities may change. The latter is associated with a change in the wealth share η in (3.2).

[11] Note that Proposition 3.2 also provides a closed-form solution for the rental yield:

$$\frac{r_s}{p_h} = \frac{1-\alpha}{1-\nu}\frac{1}{\alpha}\left[\frac{c_t}{n_t}\eta_t + \frac{\underline{c}_t}{\underline{n}_t}(1-\eta_t)\right].$$

In typical model calibrations, such as the one in Section 5, individual consumption is smooth. Similarly to Mian, Straub, and Sufi (2021), with high enough heterogeneity in saving rates, changes in the wealth distribution dominate and thus largely determine fluctuations in the discount factor.

Second, the numerator in the preference component $(1-\alpha)/\alpha$ in (3.1) has been replaced by $1-\nu$. In case of unit elastic substitution between housing and non-durable consumption, with utility of the form $u(c^\alpha s^{1-\alpha})$, in the degenerate representative agent steady-state with only entrepreneurs or savers (i.e. $\eta = 0$ or $\eta = 1$), we have $\nu = \alpha$. The factor ν therefore is largely influenced by preferences. This notwithstanding, prices are not only driven by preferences but also by the portfolio incentives of the different market participants. Indeed, the next theorem characterizes the dynamics of η and ν in full generality. Moreover, under the additional assumption 4.1 discussed in section 4, we can further rewrite the right hand sides of equations (3.3) and (3.4) in terms of the model primitives. This result, therefore, provides a full closed-form characterization of the asset prices and housing rental rate in the model.

Theorem 3.3. *The evolution of the wealth distribution, characterized by η, and relative prices, characterized by ν, are given by*

$$\frac{\dot{\eta}}{\eta(1-\eta)} = \underbrace{\frac{\underline{c}_t}{\underline{n}_t} - \frac{c_t}{n_t} + r_{st}\left(\frac{\underline{s}_t}{\underline{n}_t} - \frac{s_t}{n_t}\right)}_{\textit{Consumption difference}} + \theta_{kt} \underbrace{\left[r_{kt} - \underline{r}_{kt}\right]}_{\textit{Return difference}} \tag{3.3}$$

$$\frac{\dot{\nu}}{\nu(1-\nu)} = \underbrace{\frac{r_{st}}{p_{ht}} - \frac{a^j}{p_{kt}}}_{\textit{Yield difference}}, \tag{3.4}$$

where $a^j = a$ for $\eta > \phi_k \nu$ and $a^j = \underline{a}$ for $\eta \leq \phi_k \nu$.[12]

Wealth accumulation depends on the consumption rates and the return differential between the agents. The evolution of relative prices of capital and housing, described by ν, on the other hand, depends on the difference between the yields for the marginal agent, who invests in both assets. By the no-arbitrage condition, the total return on the two assets must equal; therefore, the difference between capital gains rates is determined by the difference in yields. The identity of the marginal agent, in turn, depends on whether the collateral constraint is binding or not.

[12]Note that with ak technology an entrepreneur may be credit constraint if and only if all entrepreneurs are credit-constrained.

Let us now consider a credit relaxation allowing constrained entrepreneurs to purchase productive capital. The capital reallocation increases aggregate productivity $A(\psi_k)$; by (3.1) rental rate increases. Consider the saver's no-arbitrage condition. In a static model, with no capital gains, the increase in rental rate yields a positive house price shock:

$$\frac{\underline{a}}{p_k} = \frac{r_s(\psi_k)\uparrow}{p_h\uparrow}.$$

Savers valuation for the business capital is given by $p_k = \underline{a}$. Given a demand shock on business capital from the entrepreneurs, savers provide fully elastic supply of capital to them, and rebalance their portfolios toward housing. These mechanics are depicted in Figure 1 and microfounded in detail in Appendix A.3.

Let us now consider a dynamic model, where the initial demand shock on business capital by the entrepreneurs also affects the price of capital and, building on the intuition from the static example, also generate a sustained house price boom, where house prices increase disproportionately compared to the price of capital:

$$\frac{\underline{a}}{p_{kt}\uparrow} + \frac{dp_{kt}}{p_{kt}} = \frac{r_{st}(\psi_{kt})\uparrow}{p_{ht}\uparrow\uparrow\uparrow} + \underbrace{\frac{dp_{ht}}{p_{ht}}}_{\text{Assume } \frac{dp_{ht}}{p_{ht}} > \frac{dp_{kt}}{p_{kt}}}.$$

If $\frac{dp_{ht}}{p_{ht}} > \frac{dp_{kt}}{p_{kt}}$, house prices are amplified even further, as the lower rental yield is now compensated by higher expected future house price appreciation.

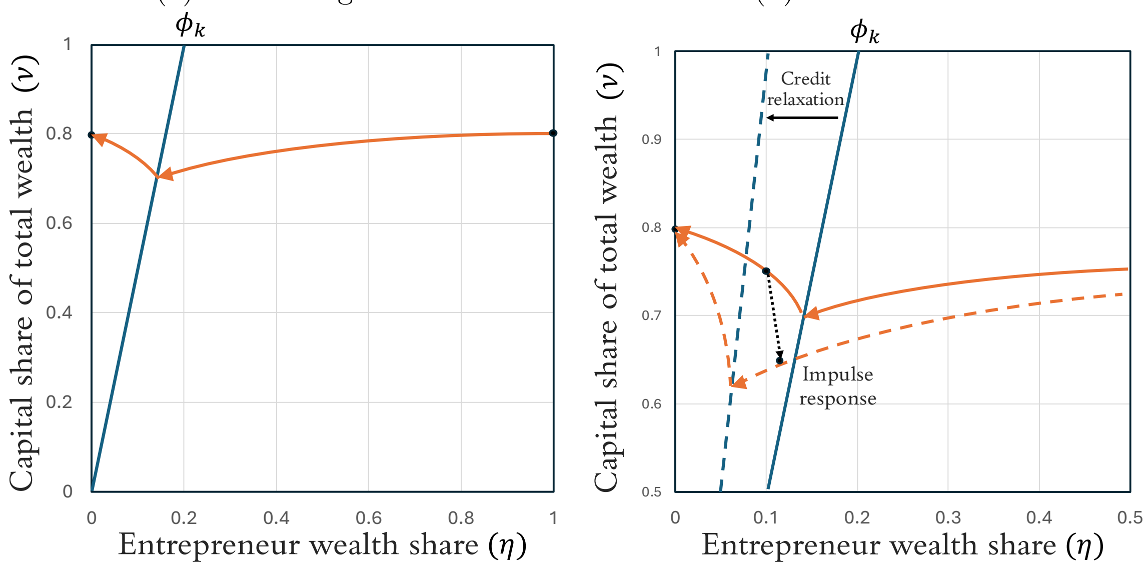
This notwithstanding, if the credit shock lifts the entrepreneurs completely off the credit constraint, the identity of the marginal agent shifts, as shown in Theorem 3.3, where the entrepreneurs are credit-constrained if and only if $\eta < \phi_k \nu$. In this case, the relevant no-arbitrage condition is that of the entrepreneurs, and the dividend yield on left hand side of the above equation is $a/p_{kt} > \underline{a}/p_{kt}$. In particular, the supply of capital becomes fully inelastic once the entrepreneurs own all the capital in the economy. To total price effect, therefore, depends on the relative magnitudes of each of these forces.

In section 4, I will provide a full characterization of the dynamics in terms of model primitives, but for now, I illustrate these dynamics in Figure 3, corresponding to the first parametric case of Proposition 4.5. I discuss this assumption in more detail in section 4 and now only note that this assumption implies that the entrepreneurs' productivity advantage dominates the effects of their lower patience. In this parametric case, the model is non-stationary and only has two degenerate steady-states corresponding to the corner cases, where either savers own all assets ($\eta = 0$) or the entrepreneurs do ($\eta = 1$). Figure 3 depicts an (η, ν) phase diagram of the model: the orange line in panel (a) shows the model saddle path, presenting the model dynamics along the transition. The credit constraint is depicted by the solid blue line, which divides the plane into the constrained region (left) and unconstrained region (right). When the dynamics cross this line, the identity of the marginal agent determining asset prices changes from the saver to the entrepreneur; the dynamics correspondingly have a point of non-differentiability, due to the discontinuous drop in the productivity of the marginal agent from a (entrepeneur productivity) to $\underline{a}$ (saver productivity).

Figure 3: Phase diagram

The figure demonstrates the model dynamics and the impulse response to an unanticipated, exogenous credit shock, under Assumption 4.1 (log utility) for parameter configuration associated with (4.5). Panel (a) shows the saddle path in a (η, ν) phase diagram. The line $\eta = \phi_k \nu$ separates the region, where the credit constraint is binding (left) from the non-binding region (right). Panel (b) considers an unanticipated, exogenous credit relaxation and depicts the impulse response, as the economy readjusts onto the new saddle path. The credit shock is represented by a rotation of the $\eta = \phi_k \nu$ line. The model is non-stationary and has two steady states corresponding to the two degenerate representative agent economies associated with $\eta = 0$ and $\eta = 1$, where either savers or entrepreneurs own all wealth in the economy, respectively.

(a) Phase diagram (b) Credit shock



Panel (b) of the figure considers an unanticipated credit shock rotating the blue line in the figure. In the figure, I have initialized the economy into the constrained region, and consider the impulse response, as the economy readjusts to the new saddle path in the unconstrained region. The relative price of housing, relative to capital price, described by ν, increases at impact, but it also keeps increasing after the shock, generating a sustained house price boom, until the dynamics bring the economy back to the constrained region, where the house price effect reverts. The entrepreneurs' higher consumption propensity leads them to liquidate their capital holdings in order to finance higher consumption. Eventually, they become credit constrained and begin to liquidate also their productive business, which generates capital deterioration, and reversal of the price mechanism.

The above results hold in full generality, where I have not specified the utility function of the model. In section 4, I will complete the analyses by such further assumptions. Before doing so, it is however worthwhile to comment the model features and their relation to the data in some more detail.

3.1 Model features and empirical relevance

Several of the model features can be mapped to empirical evidence in the existing literature.

3.1.1 Financial constraints and aggregate productivity

A large body of empirical research has shown that collateral constraints affect investments and business activity at both micro and macro levels (Chaney et al., 2012; Liu et al., 2013; Robb and Robinson, 2014; Cvijanović, 2014; Adelino et al., 2015, 2016; Schmalz et al., 2017; Bahaj et al., 2020; Doerr, 2021). Building on this empirical motivation, I follow the existing theory literature (Buera and Moll, 2015; Buera and Nicolini, 2020) and embed my model with such a mechanism that guarantees that credit conditions affect aggregate output. While I motivate this via the entrepreneurial finance setting, in essence, the model only requires that financial frictions affect aggregate output, an assumption deeply rooted in the existing literature (Whited, 2006; Buera and Shin, 2013; Moll, 2014; Midrigan and Xu, 2014; Whited and Zhao, 2021). In particular, my model makes no predictions on entry or exit of entrepreneurs.[13] For a more thorough survey of the role of borrowing constraints for aggregate output, I refer to Hopenhayn (2014).

Going beyond this general motivation, an immediate corollary of Proposition 3.1—and a testable prediction of the model—is that firms with real estate holdings (i.e.,

[13] Decker et al. (2014) document that entrepreneurship and "business dynamism," in general, have decreased in the U.S.A. over the last few decades. This observation suggests that the relevant margin for capital reallocation is not due to firm entry and exit but via the intensive margin, which is consistent with the structure of my model presented in section 2. Nevertheless, adjusting the model mechanism to work through business creation as an alternative interpretation would not be difficult.

savers) are less productive than other firms who do not own any housing assets (i.e., entrepreneurs). This matches the empirical findings of Doerr (2020), which documents a lower total factor productivity with U.S. public firms with a high share of assets in real estate.[14]

Regarding the aggregate productivity during the pre-crisis years in the U.S.A., in particular, Gomme, Ravikumar, and Rupert (2011) document that returns to business capital in the U.S.A. increased from 3.8% to 11.8% during the housing boom of Q1/2000 – Q4/2006. Returns increased throughout the early 2000s, even when excluding increasing capital gains, reaching the highest since 1975. At the macro level, the U.S. G.D.P. growth averaged 3.9% per annum from Q1/2003 to Q1/2006, peaking at 7.0% in Q3/2003, consistent with the credit supply shock associated with the private-label securitization of mortgage-backed securities driving a productivity-enhancing capital reallocation (Levitin and Wachter, 2011; Buera, Kaboski, and Shin, 2011; Buera and Shin, 2013).

Finally, I note that the model takes no stance on the exact source of the increase in capital productivity. For instance, if a low-productivity entrepreneur is willing to increase their working hours to expand their business, this could increase the aggregate productivity of capital while decreasing labor productivity. On the other hand, the mechanism is also agnostic about who benefits from the increase in productivity: the entrepreneurs themselves with higher profits, their employees with higher wages, or their customers with lower prices. Altogether, the model makes no empirical predictions on labor productivity,[15] the income[16] of any particular group of entrepreneurs

[14] Doerr (2020) argues that increasing house prices decreases aggregate productivity because lower-productivity firms, which own much real estate, benefit the most from the increased collateral capacity. My mechanism agrees with Doerr (2020) in that the firms with real estate holdings are less productive but also proposes an additional opposing force with collateral valuations helping smaller entrepreneurs to grow, potentially increasing aggregate productivity and employment in line with Adelino et al. (2015). In particular, the negative association between real estate holdings and productivity, as documented by Doerr (2020), concerns publicly listed firms with up to 50% of their assets in real estate. In contrast, my mechanism is better interpreted in the context of private start-ups with little real estate holdings (Haltiwanger, 2015; Doerr, 2021).

[15] The question of business cycle cyclicality of labor productivity is debated and depends on the period considered (Fernald and Wang, 2016). A classic textbook mechanism suggests that average labor productivity should be countercyclical since marginal labor productivity is decreasing; this implies that every additional employee hired during a boom decreases average labor productivity (Biddle, 2014). With heterogeneous workers, firms also only keep their most productive employees during a recession, whereas during booms, workers with lower labor productivity can be hired, too. In general, the question is nuanced and debated, but in any event, the evolution of average labor productivity during the boom does not provide a reliable test for my mechanism.

[16] Mian and Sufi (2009) consider the drivers of the credit expansion in the early 2000s and document a negative correlation between income and credit growth across U.S. zip codes, suggesting the credit expansion was not driven by income growth. While my mechanism agrees with their main conclusion

(Mian and Sufi, 2009), or the sectors in which the entrepreneurs operate (Mian, Sufi, and Verner, 2020, cf. section 3.1.3 below).

3.1.2 Portfolio rebalancing

The key feature of my price mechanism is that the credit relaxation pushes the marginal agents determining asset prices (i.e. savers) to rebalance their portfolios toward housing. Recent macro housing literature has established that the "tenure supply" elasticity—i.e., how housing investors' portfolio decisions respond to demand shocks—is a key determinant of house price effects (Kaplan et al., 2020; Greenwald and Guren, 2021). In the model of Kaplan et al. (2020), increased housing demand due to a credit relaxation is met by investors, who elastically sell their investment properties to new homeowners; homeownership rate increases with little price effects.

Instead of the supply of homes, my model revolves around the supply of capital. The savers supply capital elastically to the entrepreneurs, and instead of pulling out of the housing market like in Kaplan et al. (2020), they rebalance their portfolios toward housing. Indeed, while the homeownership rate did increase during the housing boom of the early 2000s in the U.S.A., so did investor activity on all levels of the housing market: (i) individual investor activity, often labeled as housing "speculation", increased dramatically on the housing market during the boom (DeFusco, Nathanson, and Zwick, 2022),[17] (ii) institutional investors increased their portfolio shares on real estate investments; for example, local and state government pension funds' average portfolio shares in U.S. real estate doubled from 3.1% in 2000 to 6.5% in 2008 (Pennacchi and Rastad, 2011), exceeding the passive effect of growing house prices and suggesting an active portfolio rebalancing towards real estate, and (iii) housing portfolio shares dramatically increased also for individual households, but with substantial heterogeneity in portfolio evolutions across different types of households and investors (Martinez-Toledano, 2020).

Altogether, investors' portfolio choices are at the heart of the question of how credit conditions affect house prices (Kaplan et al., 2020; Greenwald and Guren, 2021). While the empirical evidence on the topic is still limited, if anything the

that the credit expansion was due to a credit supply shock (Levitin and Wachter, 2011), I also note that my mechanism emphasizing the increase in aggregate productivity is not inconsistent with negative income growth, if the benefits of the productivity growth are recouped by firms' customers or employees living outside the zip code. As I document in Figure 2 and discuss in Appendix B, business loan originations and measures of business activity are correlated at the county level.

[17] See also Gargano and Giacoletti (2022) on retirees "reaching for income" by investing in housing during low interest rate periods, when alternative assets produce relatively low yields.

evidence seems to point to the housing market attracting more investor activity during the booms, consistent with my theory of portfolio rebalancing.

3.1.3 Productivity or demand?

An interesting feature of the model is the inherent identification challenge it embeds: housing rents and prices increase because of the demand generated by the increase in productivity. In other words, the model predicts a strong causal demand effect on house prices, consistent with much of the empirical literature emphasizing the demand channel in explaining housing booms. Nevertheless, the root cause underlying this demand effect is the increase in aggregate productivity, which also has strong empirical support, as documented above.

Solving this identification challenge empirically is well beyond the scope of this article, but it highlights the highly nuanced nature of the problem, and can potentially inform future empirical research on the topic. I also note that this problem has been explicitly tackled in the existing literature by Mian, Sufi, and Verner (2020). They study whether credit supply expansions operate by increasing the economy's "productive capacity" or via "household demand". Their analysis relies on the work of Bahadir and Gumus (2016), who argue that demand shocks are inflationary, whereas productivity shocks are not. Therefore, demand shocks should differentially affect the non-tradeable sector, which Mian et al. (2020) then confirm using natural experiments in the U.S.A. in the 1980s.

However, it should be entirely plausible that the effects of collateralized lending primarily operate on the non-tradeable sector, and especially in regards to small businesses, it seems difficult to rule out the "productive capacity" channel by merely documenting differences in tradable vs. non-tradeable economic activity. In particular, another strand of the empirical and quantitative literature has documented that financial frictions may substantially affect aggregate productivity (Buera, Kaboski, and Shin, 2011; Whited and Zhao, 2021). If credit expansions operate primarily via the household demand channel, it raises the question, why does the quantitative macro literature assign such substantial productivity effects on financial misallocation.

4 House price boom-bust in general equilibrium

As discussed in section 3, the price mechanism consists of two parts: (i) an exogenous relaxation of a collateral constraint leads to a large increase in house prices at the impact of the shock; (ii) the positive impulse response in house prices is then succeeded by a sustained house price boom. In this section, I will discuss these results in a fully-specified model, and also explain, how the model generates a number of additional findings: (i) large consumption elasticity from a house price shock (Berger et al., 2017), (ii) house price momentum (Guren, 2018; Case and Shiller, 1988), (iIi) decreasing interest rates during a sustained boom (Mian et al., 2021), and (iv) simultaneous discontinuous drop in the interest rate ("flight-to-safety") and decreasing output (Guerrieri and Lorenzoni, 2017).

Similarly to Greenwald (2018), I assume logarithmic utility.[18]

Assumption 4.1. All agents derive utility from non-durable consumption c and housing services s with utility function

$$u(c,s) = \alpha \log c + (1-\alpha)\log s, \tag{4.1}$$

where $\alpha \in (0,1]$.

Apart from analytic convenience, this utility specification has two main reasons. First, Piazzesi, Schneider, and Tuzel (2007), Davis and Ortalo-Magné (2011) and Aguiar and Hurst (2013) document that the elasticity of substitution between housing and non-durable consumption is close to 1. This argues for a utility specification of the form $u(c^\alpha s^{1-\alpha})$. Moreover, the model has no uncertainty and thus the logarithmic form corresponds to setting intertemporal elasticity of substitution to 1, which does not seem far-fetched relative to the reality.

For demonstrating the dynamics, logarithmic utility is convenient also because then the agents are PIH[19] consumers. We have the following Lemma.

Lemma 4.2. *Suppose all agents have logarithmic utility* (4.1)*. Then*

$$\frac{c_t}{n_t} = \alpha\rho \quad \textit{and} \quad \frac{\underline{c}_t}{\underline{n}_t} = \alpha\underline{\rho} \quad \textit{as well as} \quad \frac{r_{st}s_t}{n_t} = (1-\alpha)\rho \quad \textit{and} \quad \frac{r_{st}\underline{s}_t}{\underline{n}_t} = (1-\alpha)\underline{\rho}$$

[18] Greenwald (2018) considers logarithmic utility over non-durable consumption and housing services, but contrary to my model, where output is produced by capital only, his utility specification includes a third term for disutility of working, as he considers a model with labor-only production.

[19] Permenant income hypothesis.

for all t. Moreover, $\nu = \alpha$, if $\eta = 0$ or $\eta = 1$.

This result has several important consequences. First, similarly to Berger, Guerrieri, Lorenzoni, and Vavra (2017), it follows that consumption response to a house price shock is given by

$$\frac{\partial c_t}{\partial p_{ht}/p_{ht}} = \text{MPC} \cdot p_{ht} h_t, \tag{4.2}$$

when keeping all other prices and portfolios constant, but allowing net worth to evolve with the change in house prices: the consumption response for a house price shock by PIH agents is purely due to a wealth effect. In particular, given large increases in house prices, the model produces large consumption responses as documented empirically by Mian, Rao, and Sufi (2013).

Moreover, as discussed in the following Proposition, for highly levered agents, the consumption responses are relatively higher; this corresponds to empirical findings by Mian and Sufi (2014), who document strong and significant effect of leverage on marginal propensity to consume out of housing wealth.

Proposition 4.3. *Suppose Assumption 4.1 holds. Keeping all other prices and the agent's portfolio constant, but allowing consumption to react to a change in net worth, the consumption elasticities for capital price shocks are then given by the corresponding portfolio shares as*

$$\frac{\partial c_t/c_t}{\partial p_{ht}/p_{ht}} = \theta_{ht} \quad \textit{and} \quad \frac{\partial c_t/c_t}{\partial p_{kt}/p_{kt}} = \theta_{kt}. \tag{4.3}$$

Moreover, the aggregate consumption response is given by

$$\frac{\partial C_t/C_t}{\partial p_t/p_t} = \eta_t^{APC} \frac{\partial c_t/c_t}{\partial p_t/p_t} + (1 - \eta_t^{APC}) \frac{\partial \underline{c}_t/\underline{c}_t}{\partial p_t/p_t},$$

where

$$\eta_t^{APC} = \frac{\eta_t APC}{\eta_t APC + (1 - \eta_t)\underline{APC}} \tag{4.4}$$

is the average consumption propensity weighted entrepreneur wealth share.

Another important implication of Lemma A.13 is that the rental yield may be written as

$$\frac{r_s}{p_h} = \frac{1-\alpha}{1-\nu}\left[\rho\eta + \underline{\rho}(1-\eta)\right].$$

In the unconstrained region $\dot{\nu} < 0$ and $\dot{\eta} < 0$ implying that the rental yield is therefore necessarily decreasing: price-to-rent ratio increases during the boom. A similar result holds for the risk-free interest rate. Indeed, we can characterize its behavior via the following Proposition.

Proposition 4.4. *Suppose Assumption 4.1 holds. Then in the unconstrained region, characterized by $\eta \geq \phi_k \nu$, the risk-free rate is given by*

$$r_{ft} = r_f(\eta_t) = \rho\eta_t^{APC} + \underline{\rho}(1 - \eta_t^{APC}),$$

where η_t^{APC} is as in (4.4).

The risk-free interest rate is given by a properly weighted average of agents time discount rates. This is similar to Mian et al. (2021), where interest rate is driven by shifts in the wealth distribution of agents' with different saving propensities.

Recall that house prices are by Proposition 3.2 given by

$$p_{ht} = \frac{r_{st} + \dot{p}_{ht}}{r_{ft}} = \frac{A(\psi_k)}{\alpha\big[\rho\eta + \underline{\rho}(1-\eta)\big]}\frac{K}{H}(1-\nu),$$

where

$$\psi_k = \begin{cases} 1, & \text{if } \eta \geq \phi_k\nu, \\ \frac{\eta}{\phi_k\nu}, & \text{if } \eta < \phi_k\nu. \end{cases}$$

For a large enough credit relaxation, i.e. reduction in the value of ϕ_k, entrepreneurs are lift off their credit constraints, after which $\psi_k = 1$ and $A(\psi_k) = a$. The shock is followed by a sustained house price boom, which is partially driven by decreasing interest rates: the risk free rate r_{ft} as well as the above discount factor $\rho\eta + \underline{\rho}(1-\eta)$ are decreasing, since wealth is being accumulated by the savers, who have high saving rates. High desired savings increase the prices of capital.

The exact long-run model dynamics depend on the model parametrization characterized in the following proposition.

Proposition 4.5. *The entrepreneur wealth share evolves with $\dot{\eta} > 0$ in the region, where $\eta > \phi_k\nu$. For $\eta \leq \phi_k\nu$, there are three possibilities:*

1. *Suppose*

$$\frac{a}{\underline{a}} - 1 < \phi_k\left[\frac{\rho}{\underline{\rho}} - 1\right]. \tag{4.5}$$

Then the equilibrium entrepreneur wealth share η satisfies $\dot{\eta} \leq 0$ for all t, and the only steady-states are at $\eta = 0$ and $\eta = 1$ with $\nu = \alpha$. Only the steady-state at $\eta = 0$ is dynamically stable.

2. *Suppose*

$$\frac{a}{\underline{a}} - 1 > \phi_k\left[\frac{\rho}{\underline{\rho}} - 1\right].$$

Then, the model has an interior steady-state at

$$\eta_{ss} = \frac{\underline{\rho}\alpha\phi_k}{\underline{\rho} + (\rho - \underline{\rho})\phi_k(1-\alpha)} \quad \text{with} \quad \nu_{ss} = \frac{\eta_{ss}}{\phi_k}. \tag{4.6}$$

with dynamically stable dynamics both for $\eta < \phi_k \nu$ *and* $\eta > \phi_k \nu$.

3. *Suppose*

$$\frac{a}{\underline{a}} - 1 = \phi_k \left[\frac{\rho}{\underline{\rho}} - 1\right]. \tag{4.7}$$

Then, the loci $\dot{\eta} = 0$ *and* $\dot{\nu} = 0$ *coincide in the region, where* $\eta \leq \phi_k \nu$, *constituting to a continuum of steady-states.*

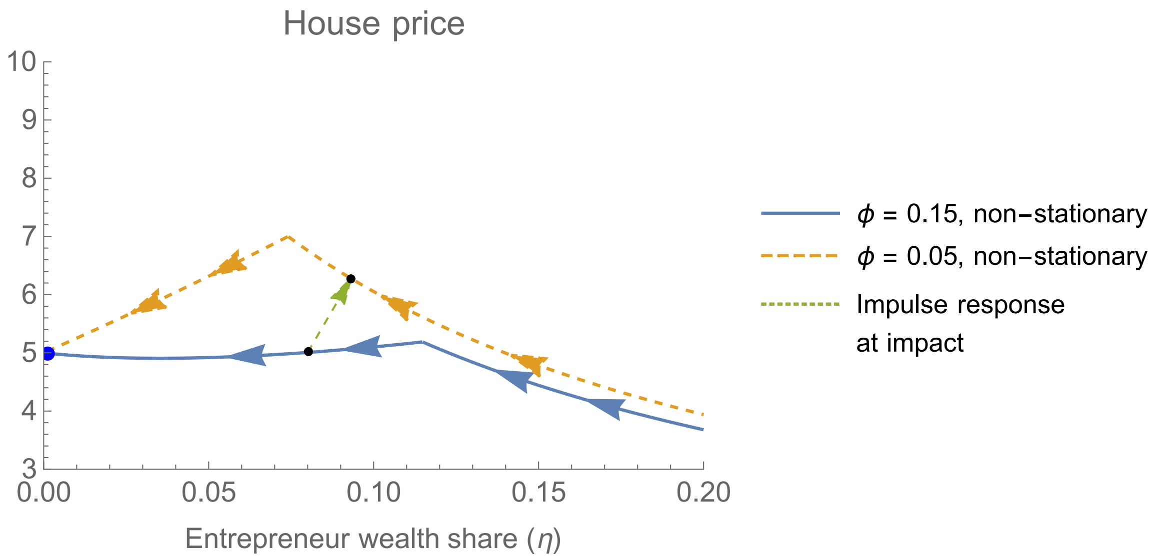
My model calibration concerns the first case in Proposition 4.5. Kumhof, Rancière, and Winant (2015) argue that the housing crisis was partially amplified due to a sequence of shocks, which increased income inequality and made the economy more vulnerable to large business fluctuations. While my mechanism is different from theirs, the empirical observation of increasing income and wealth inequality corresponds to the first case of Proposition 4.5 with non-stationary dynamics. For a phase diagram representing the model dynamics, I refer to Figure 3 discussed in section 3.

4.1 LTV collateral relaxation with an endogenous bust

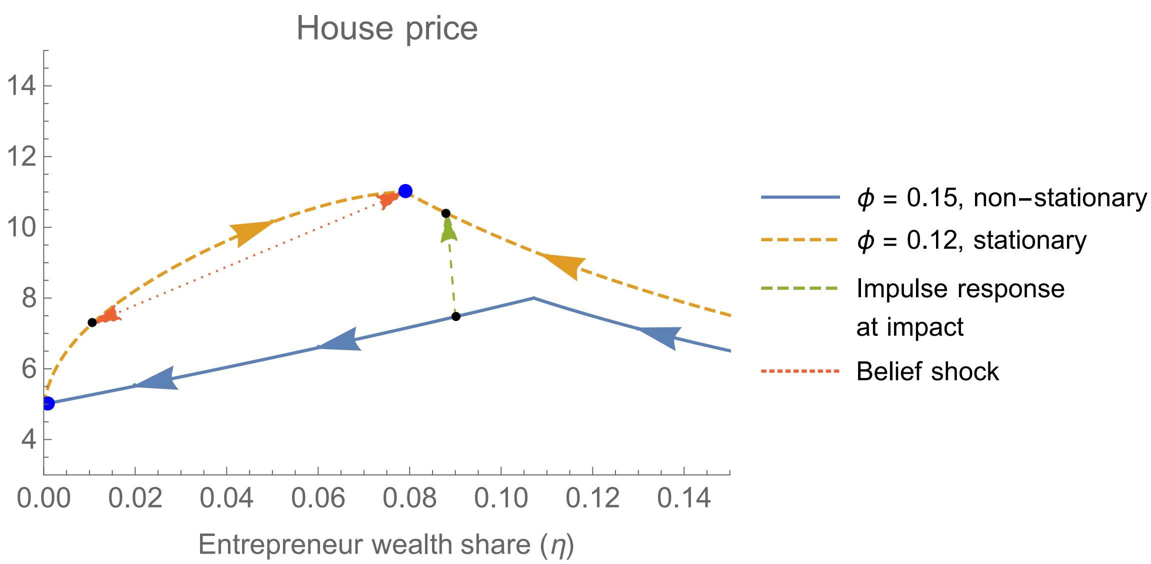
The main experiment I consider is an unanticipated shock decreasing the value of the leverage parameter ϕ_k. I begin by showing the house price dynamics for the case (a) of Proposition 4.5, which is depicted in panel (a) of Figure 4 for two different parameterizations: $\phi_k = 0.15$ and $\phi_k = 0.05$, keeping other parameters unaltered across these specifications. The model has two steady-states: $\eta = 0$ and $\eta = 1$, and I concentrate on $\eta = 0$, since it has a dynamically stable saddle path. The steady state is depicted as a blue dot.

The transition paths have a point of non-differentiability, when the path crosses the line $\eta = \phi_k \nu$ in Figure 3. For $\eta > \phi_k \nu$ all of the productive capital is owned by the entrepreneurs; at the boundary $\eta = \phi_k \nu$, they begin to sell their productive capital to the savers at a positive rate: this produces a kink in the pricing schedules as well as a discontinuity for the risk-free interest rate. In particular, the collateral constraint is binding to the left of the kink in panel (a) of Figure 4. When an entrepreneur hits the collateral constraint, she sells her capital to the less productive savers at a positive rate. This deteriorates resource allocation and produces non-monotonic dynamics.

Figure 4: House price impulse response: two cases



(a) The picture depicts equilibrium price schedules as a function of the entrepreneur wealth share across the state space for $\eta \in [0, 0.2]$ corresponding to the case (a) of Proposition 4.5. For $\phi_k = 0.15$ and $\phi_k = 0.05$ the transitions are depicted in solid blue and dashed yellow lines, respectively. The economy is initialized at $\eta = 0.08$ and the impulse response at the impact of the shock is shown (dashed green), given an unanticipated exogenous shift in the value of ϕ_k from $\phi_k = 0.15$ to $\phi_k = 0.05$.



(b) The picture depicts equilibrium price schedules as a function of the entrepreneur wealth share across the state space for $\eta \in [0, 0.2]$. For $\phi_k = 0.15$ the non-stationary transition is shown (solid blue) corresponding to case (a) of Proposition 4.5. The economy is parametrized to produce the dynamics of case (b) of Proposition 4.5, given an exogenous shift in the value of $\phi_k = 0.15 \to 0.12$. The corresponding price schedule following the shock is depicted (dashed yellow) together with the impulse response (dashed green) for an initial state $\eta_o = 0.09$. The steady state following the LTV shock is vulnerable to a belief shock affecting the house price (double-sided, dotted red).

With both parameterizations depicted in panel (a) of Figure 4, the entrepreneurs thus eventually sell off all their housing stock and only hold productive capital in the long term. This is because of their lower saving rates compared to savers; entrepreneurs' relative wealth, and hence asset holdings, must decrease. Due to the entrepreneurs higher production efficiency, they always choose to first sell off their housing holdings before liquidating their income-generating capital. Consequently, only the productive capital may have any collateral value.

An unanticipated LTV shock affects the equilibrium quantities and, in particular, it moves the house price schedule. Concentrating first on panel (a), the original price schedule for $\phi_k = 0.15$ is depicted in solid blue line, whereas the new price schedule after an unanticipated shock $\phi_k = 0.15 \to 0.05$ is shown in a dashed yellow line. The initial impulse response at the impact of the shock is depicted with a dashed green arrow, given an initialization of the economy at $\eta = 0.08$.

The total price effect consists of three separate components. First, before the shock some of the productive capital was held by the savers, who are less efficient producers: the pre-shock wealth share located left of the point of non-differentiability, whereas after the shock, the economy moves to the right of this point in Figure 4. This reflects a change in the resource allocation increasing output and house prices. Secondly, even in the region, where the collateral constraint is not binding, the price schedule moves upwards. This is due to changes in the relative value of capital and housing, taking into account the forces explained in Section 3; most notably, the relative demand for housing and capital associated with the agents' portfolio decisions. Third, the shock produces a shift in the wealth distribution, since the agents have different portfolios. This affects the discount rate used for pricing assets. At the impact of the shock, this shift may mitigate the price effect, but in the long term the effect is necessarily overturned. In Section 5 (cf. Figure 6) each of these forces is studied quantitatively.

In panel (b) of Figure 4, I consider another parametrization, where the pre-shock dynamics are given by the non-stationary saddle path associated with case (a) of Proposition 4.5, but the exogenous relaxation of the LTV constraint from $\phi_k = 0.15$ to $\phi_k = 0.12$ moves the economy to case (b) of the Proposition. After the shock, the economy moves towards an interior steady state characterized by (4.6). The new steady-state as well as the associated transition path is, however, still vulnerable to belief shocks.

Starting from the steady-state, the model includes two equilibria corresponding to two different price paths: (i) one, where nothing happens and the economy stays at the steady-state, and (ii) another one, where the economy moves to another point at the transition path, leading to a house price crash. These equilibria are associated with different beliefs about the aggregate state evolution. In particular, a coordinated change in the agents' beliefs about the proper price level may relocate the economy to a different point of the state space, associated with different resource allocation and different prices. Belief shocks may thus have large price effects, for instance, producing a market crash. The strategic complementarity producing this multiplicity is discussed in more detail in section A.4.2, leveraging the simplified setting of the static model. We omit the discussion here, since our baseline calibration relates to the case depicted in panel (a) of Figure 4.

5 Quantitative analysis

I now consider the quantitative implications of my model in greater detail and for this purpose, I introduce some extensions to the model set-up introduced in Section 2. In particular, I allow for investments such that for each individual j, her capital stock evolves according to

$$\frac{dk_t^j}{k_t^j} = \Phi_k(i_{kt}^j) - \delta_k \, dt$$

and housing stock according to

$$\frac{dh_t^j}{h_t^j} = \Phi_h(i_{ht}^j) - \delta_h \, dt,$$

where i_{kt}^j and i_{ht}^j are the capital and housing investment rates, respectively, δ_k and δ_h are the capital and housing depreciation rates, and

$$\Phi_k(i) = \frac{1}{\kappa_k} \log(1 + \kappa_k i) \quad \text{and} \quad \Phi_h(i) = \frac{1}{\kappa_h} \log(1 + \kappa_h i)$$

are the adjustment cost functions for capital and housing investments, respectively. Capital investments, $i_{kt}, \underline{i}_{kt}$ for entrepreneurs and savers, respectively, are paid for by production of non-durable consumption goods, given the market clearing equation

$$c_t + \underline{c}_t = (a - i_{kt})k_t + (\underline{a} - \underline{i}_{kt})\underline{k}_t.$$

Similarly, I assume that housing investments make some of the housing stock unusable

$$s_t + \underline{s}_t = (1 - i_{ht})h_t + (1 - \underline{i}_{ht})\underline{h}_t.$$

The interpretation of this setup is that in order to renovate a house, or redevelop the building, the house/lot cannot be used for rental purposes during the process and, therefore, it won't generate rental revenues for its owner.

With this specification the return on holding capital is given by

$$r_{kt}^{j} \equiv \max_{i_{kt}} \quad \underbrace{\frac{a^j - i_{kt}}{p_{kt}}}_{\text{Dividend yield}} + \underbrace{\Phi_k(i_{kt}) - \delta_k}_{\text{Net investments}} + \underbrace{\frac{dp_{kt}}{p_{kt}}}_{\substack{\text{Capital price}\\ \text{appreciation}}} .$$

and the return for holding housing is similarly given by

$$r_{ht} \equiv \max_{i_{ht}} \quad \underbrace{\frac{r_{st}[1 - i_{ht}]}{p_{ht}}}_{\text{Rental yield}} + \underbrace{\Phi_h(i_{ht}) - \delta_h}_{\text{Net investments}} + \underbrace{\frac{dp_{ht}}{p_{ht}}}_{\substack{\text{House price}\\ \text{appreciation}}} .$$

It follows that the supply schedule for investments of both types of capital take the forms

$$p_k = 1 + \kappa_k i_k \quad \text{and} \quad \frac{p_h}{r_s} = 1 + \kappa_h i_h. \tag{5.1}$$

In Table 4 I demonstrate that the latter model for price-to-rent ratios and housing investments is of empirical relevance, and produces a good fit for price and investment data. This observation motivates the use of an adjustment cost model for considering the effect of housing supply on house prices.

The response of housing supply to one percentage increase in housing price-to-rent ratio is now governed by the parameter κ_h as

$$\frac{d\Phi_h(i_h)}{\frac{d(p_h/r_s)}{p_h/r_s}} = \Phi_h'(i_h)\frac{di_h}{\frac{d(p_h/r_s)}{p_h/r_s}} = \frac{1}{\kappa_h}.$$

Much of the existing literature has assumed fixed housing supply, and this set-up is built to relax this assumption, without losing the model tractability.

5.1 Calibration

There are four crucial sets of parameters affecting my mechanism: (i) κ_h determines the housing supply elasticity and directly affects house price responses for economic shocks; (ii) the ratios $a/\underline{a}$ and $\rho/\underline{\rho}$ determine the extent of price amplification within the constrained region; see also Proposition 4.5; (iii) the evolution of K_t relative to H_t affects house prices and is governed by the investment technology parameters κ_k, δ_k

and δ_h, in addition to κ_h; and (iv) the extent of the credit liberalization, which is the main experiment I consider, and is governed by the pre- and post-shock levels of ϕ_k. The parameter values I consider are shown in Table 1.

Table 1: Parameter values: baseline calibration

Parameter	Name	Value	Internal	Source/Target
		Experiment		
Capital LTV constraint	ϕ_k	$0.27 \to 0.18$	N	Shift in LTV ratios
		Parameter values		
Housing preferences	$1-\alpha$	0.183	N	Housing exp. share
saver time discount	$\underline{\rho}$	0.01	N	High saving propensity
Entrepreneur time discount	ρ	0.1	N	Low saving propensity
Entrepreneur productivity	a	1.23	N	Fagereng et al. (2020)
saver productivity	$\underline{a}$	1.0	N	Normalization
Housing adjustment cost	κ_h	1407	N	See Table 4
Capital adjustment cost	κ_k	17.1	Y	See text
Housing depreciation rate	δ_h	0.0	N	See text
Capital depreciation rate	δ_k	0.153	Y	Fraumeni (1997)

I use the linear relation in the second part of (5.1) to estimate κ_h. I construct price-to-rent ratios for 2014 – 2020 using rental and house price data from Zillow; unfortunately, rental data is available only starting 2014. I do not have data on repairs and maintenance, and I will exclude this component of housing investments and correspondingly set $\delta_h = 0$. Housing investment rate is measured by the ratio of real private residential investments and the value of the housing stock. I consider both contemporaneous and lagged effect of investments on price-to-rent ratios. The regressions are reported in Table 4.

The best explanatory power with $R^2 = 0.870$ is obtained with 15 months lag and coefficient estimate of 1664 in quarterly data. When forcing the regression intercept to 1, as indicated in (5.1), the quarterly coefficient estimates are relatively robust, slightly below 6000, independent of the lag. Using this specification arising from the model, I calibrate the value of housing supply elasticity at $\kappa_h = 1407$, corresponding to quarterly value of 5,628 in our annual model calibration and to R^2 of 0.996.[20]

The savers' productivity parameter is normalized to $\underline{a} = 1.0$. This is without

[20]Producing the house price boom is robust for a more conservative calibration of $\kappa_h = 416$ corresponding to the 15 months lagged coefficient in Column (3) of Table 4. That being said, the calibrated housing supply elasticity is lower than that in Kaplan et al. (2020), who use the 30-year elasticity reported by Saiz (2010), who studies the effect of geography on long-run housing construction. My calibration is better interpreted to consider the short-run price responses.

loss of generality and corresponds to fixing the units of measurement for productive capital. I interpret the savers as bondholders, corresponding to pension and insurance funds as well as high net-worth individuals in the data. I use average returns on wealth for business owners (7.87%; high-achiever entrepreneurs) and for individuals in the top decile of the net worth distribution (6.39%) from Fagereng, Guiso, Malacrino, and Pistaferri (2020) to calibrate the productivity heterogeneity at $a = 1.23$. This represents a modest degree of heterogeneity in the agents' productivity, reflecting differences in the agents' ability for value-adding production with the assets (wealth) they own.

The time discount rates $\rho = 0.10$ and $\underline{\rho} = 0.01$ are chosen to produce ten-fold differences in marginal consumption propensities. Misra and Surico (2014) document that approximately half of the population have marginal consumption propensities close to zero, whereas another 20% consume over 50% out of unanticipated transitory income. My calibration produces substantial, but conservative heterogeneity in consumption propensities relative to observed extremes in the data, and yields pre-shock risk-free interest rate of 1.9%.

In order to concentrate on my house price mechanism, I shut down outside forces for economic growth, which might also affect house prices. Hence, the investment parameters $\kappa_k = 17.2$ and $\delta_k = 0.153$ are chosen within reasonable estimates provided in the literature to match zero growth rate in output.[21] This conservative growth is chosen to make sure that the house price effect during the boom is not driven merely by a large increase in output, reflected in a potential increase in K_t/H_t, which is ruled off by this calibration.[22]

Before the crisis most business loans were originated at LTVs close to 0.70 (Titman, Tompaidis, and Tsyplakov, 2005; Benmelech, Garmaise, and Moskowitz, 2005). I calibrate the LTV shock using the maximal commercial mortgage LTVs of approximately 0.73 in 2000 and 0.82 in 2004, reported by . These correspond to $\phi_k = 0.27$ and $\phi_k = 0.18$ before and after the credit shock, respectively. Note that mortgages constitute to a substantial part of small business loans, and calibrating the LTV shock according to commercial mortgages, therefore, seems appropriate.[23]

Finally, the preference parameter α is matched to the expenditure share of housing

[21]See Hall (2004) for estimates of κ_k and Fraumeni (1997) for BEA estimates for capital depreciation.

[22]See the formula for p_h in Proposition 3.2

[23]For instance, in Q1/2002, the total value of U.S. proprietorship and partnership mortgages was \$1.46 trillions, constituting 76% of their total debt holdings, whereas mortgages added up to 35 % of all debt for U.S. small businesses of less than 500 employees, which constitute to 99% of all U.S. firms and majority of total U.S. employment.

services: housing and utilities expenditure has been relatively stable at 18.3% of total GDP at least for the past 40 years.[24] The initial ratio of capital and housing stocks is normalized to $K_o/H_o = 1$, and the initial entrepreneur wealth share is set to $\eta_o = 0.19$ to match the 4-year duration of the boom from the credit relaxation in Q1/2003 lasting until Q1/2007.[25] Kaplan and Violante (2014) document that approximately one sixth of U.S. households are wealthy hand-to-mouth investors, who consume most of unexpected transitory income. The entrepreneurs constitute to such entrepreneurs in the data as they choose to use high leverage to finance their highly productive business projects, and the pre-shock calibration of η_o seems to be in line with this literature.

5.2 Results: impulse responses for an LTV relaxation

As shown in Figures 5 and 7, the credit relaxation leads to 52.9% increase in house prices, even though output increases only by 5.1% during the boom (when excluding any baseline growth in output, which is set to 0% before the shock). This more than explains the 40.0% house price increase between Q1/2003 and Q1/2007, during the "house price bubble"; Levitin and Wachter (2011) document that the "bubble" really formed only after 2003, as the preceding house price growth in the early 2000s was supported by increasing rents, and the price-to-rent ratios began to increase only later in 2003 − 2004.

At impact of the credit shock house prices are increased by 36.8% together with a similar change in the price-to-rent ratio. Entrepreneurs are lift off their credit constraints and they use the additional collateral capacity to purchase more productive capital and to finance higher consumption during a sustained boom lasting for 4 years, after which they again hit the constraint and prices begin to decrease.

In Figure 6, I show, how the price impact is decomposed to the different components of Proposition 3.2: (i) increase in average productivity of capital, (ii) increase in the stock of productive capital relative to that of housing, (iii) change in the discount rate, and (iv) the *portfolio channel*.[26] Panels (a) and (b), respectively, discuss the price impact relative to the initial state and to the counterfactual price evolution; note that the model is not in steady-state at the time of the shock, since the interior

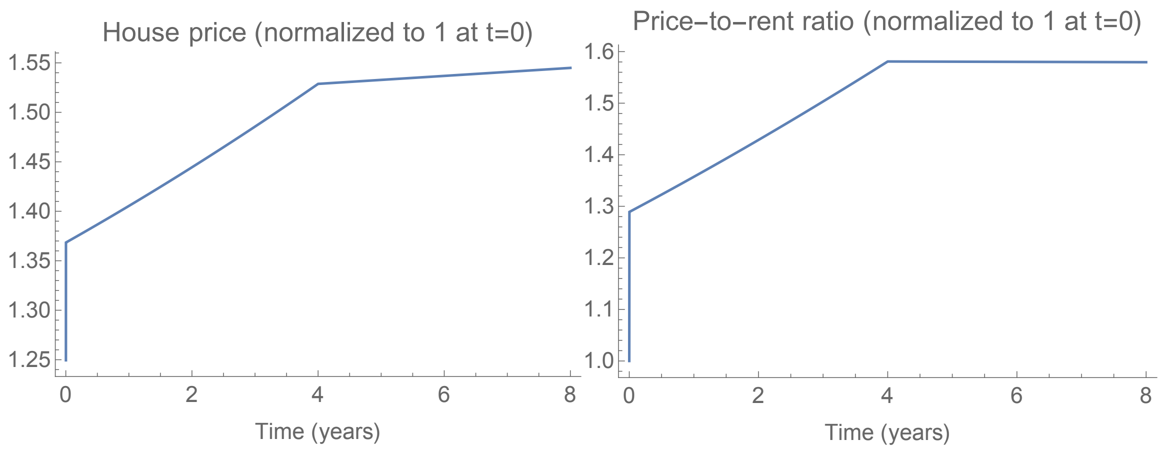
[24]Source: Personal Consumption Expenditures on Housing and utilities [DHUTRC1Q027SBEA] divided by total Personal Consumption Expenditures [PCEC], Federal Reserve Back of St. Louis.

[25]For timing the bubble see Levitin and Wachter (2011).

[26]For this quantitative analysis, Proposition 3.2 is extended to include capital investments; see Appendix D.

Figure 5: Impulse responses for house price and price-to-rent ratio.

The shock lifts the entrepreneurs off their credit constraint; a non-differentiable change in the impulse responses occurs later, when the entrepreneurs hit the constraint again.



of the state space does not have steady-states, as the model is non-stationary.

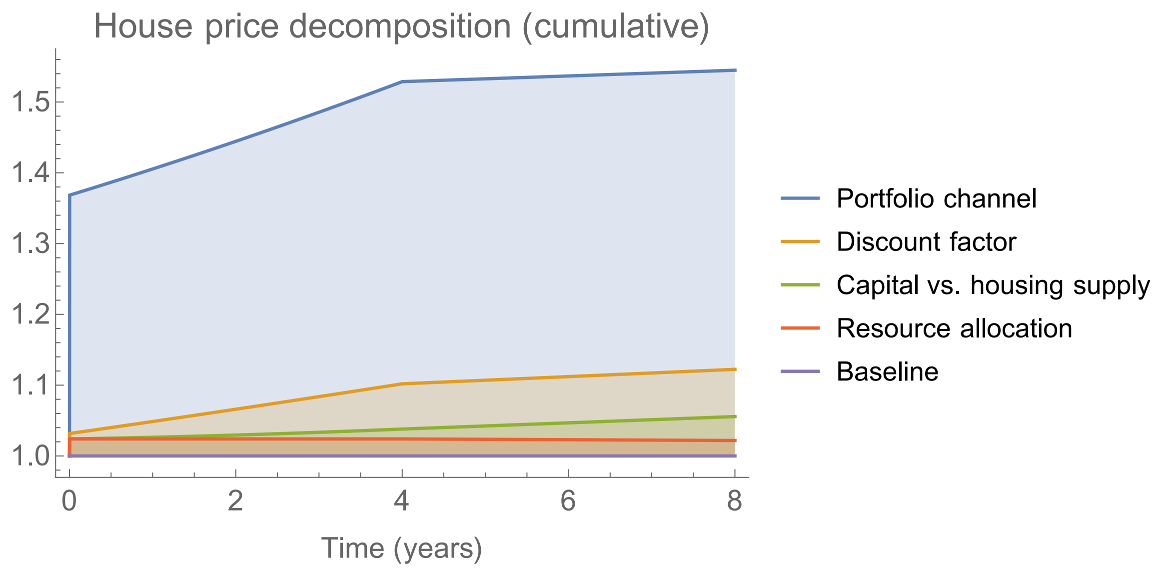
The components (i) and (ii) above also affect the price of productive capital; Figure 7 shows that the price increase in productive capital is substantially smaller compared to house prices. The large house price effect relative to price of capital is due to two reasons, both associated with the portfolio channel of (iv) above.

On one hand, the savers are willing to sell their capital to the entrepreneurs due to their differences in productivity; for housing investments savers remain marginal, and the demand for housing versus capital relative to the supply of each, therefore remains high: savers rebalance their portfolio towards housing, creating a housing boom. On the other hand, the entrepreneurs know that they will not hold the capital forever, but only for a few years during the boom, after which they prefer to liquidate it for financing higher future consumption. After the boom, capital is bought back by the savers, who are not willing to pay a high price for it due to their lower productivity. This affects the entrepreneurs' valuation for capital them being forward-looking.

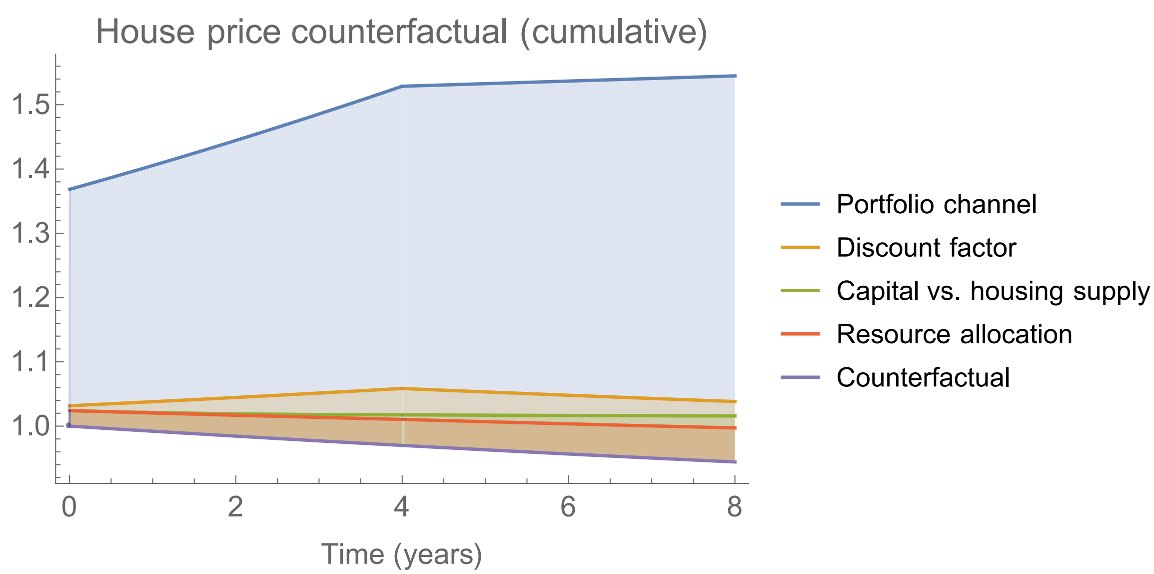
Once this liquidation begins, output decreases. House price growth flattens as savers' demand for housing is lowered, since now they are also marginal for capital investments. As the growth rates for capital and house prices decrease, the returns on these assets decrease, as well. Risk-free interest rate drops in order to balance the returns across different assets.

Note that the interest rate evolution follows the observed dynamics: during the PLS market expansion in 2003 – 2004, the long-running decrease in interest rates stopped, as the 30-year mortgage rate increased from 5.2% in June 2003 to 6.4% by

Figure 6: House price decomposition



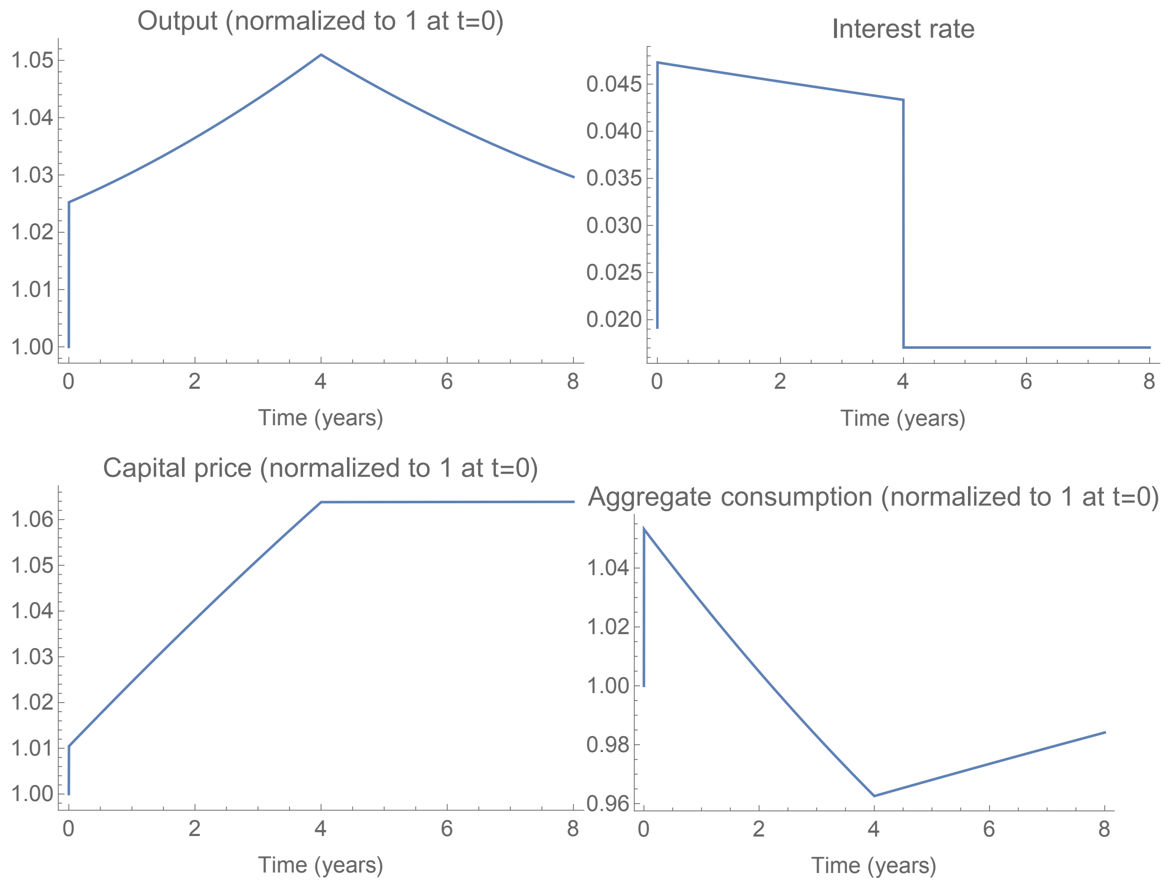
(a) The picture depicts the house price impulse response for a credit shock $\phi_k = 0.27 \to 0.18$, decomposing the price effect to its components according to Proposition 3.2.



(b) The picture depicts the house price impulse response for a credit shock $\phi_k = 0.27 \to 0.18$, decomposing the price effect to its components according to Proposition 3.2, relative to the counterfactual price evolution, where the credit shock never took place.

Figure 7: Output, interest rate, capital price and aggregate consumption.

The credit relaxation creates a strong boom in the affected sectors. Once the entrepreneurs hit the credit constraint, after being lift off the constraint at the impact of the shock, a decrease in the growth rate of real output is accompanied with a discontinuous downward jump in the real estate rate, resembling a credit crunch. Aggregate consumption increases together with the increased output at the impact of the shock, but the effect is dampened in the long run.



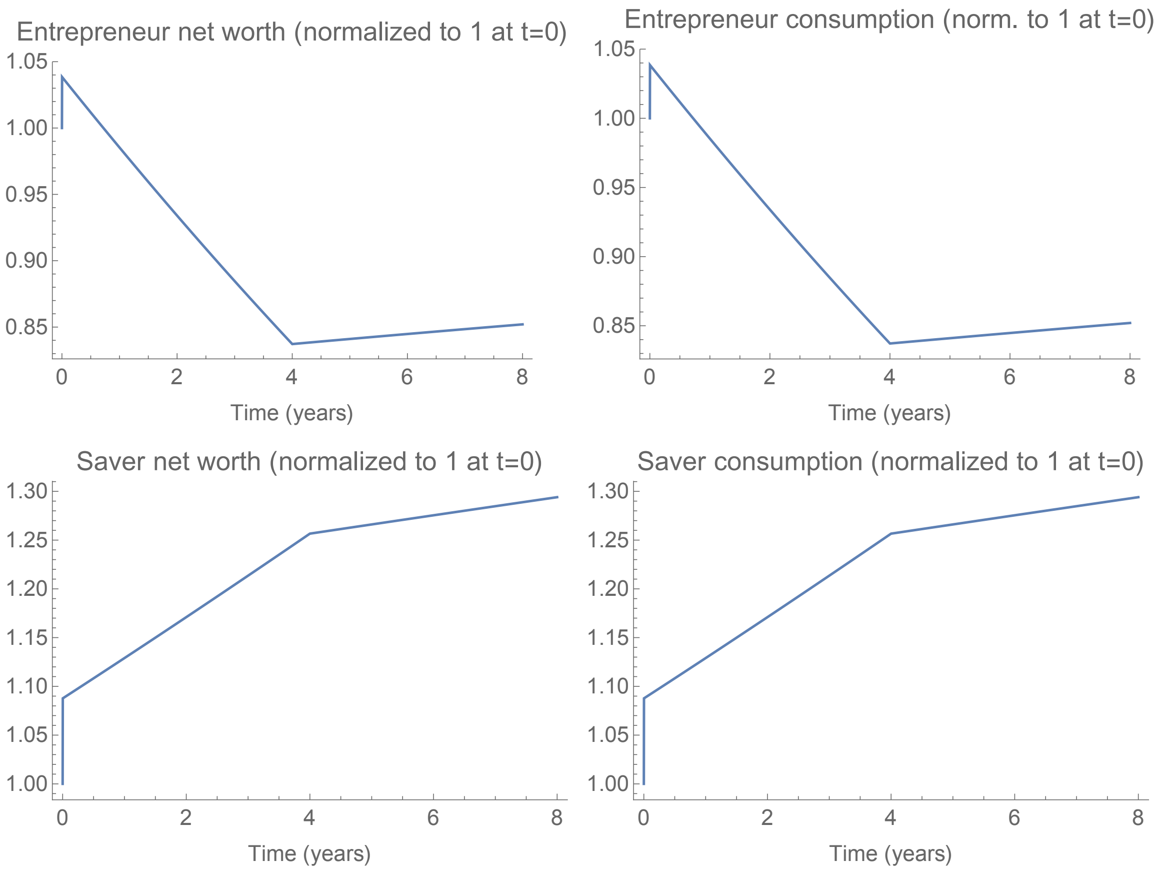
August of the same year. Afterwards, the rates remained mostly flat until the crisis, when they dropped again from 6.4% in October 2008 to 4.8% in April 2009.

In conclusion, the bust in the model is created endogenously four years after the original credit supply shock. The simultaneous drop in output and real interest rate resembles a credit crunch studied by Guerrieri and Lorenzoni (2017), who produce similar dynamics as a consequence of an exogenous credit tightening, whereas in my model the credit crunch is an endogenous outcome of the preceding credit liberalization four years earlier. Note that the price of capital (cf. Figure 7), which is the collateral in the economy, never decreases; therefore, the difference between short- and long-term debt is irrelevant for the collateral constraint.

The credit shock also produces a consumption boom, most notably for the home-

Figure 8: Consumption and net worth

Impulse responses for aggregate, saver and entrepreneur consumption as well as for entrepreneur net worth. The credit relaxation creates a consumption boom, especially for the home owners (savers), who enjoy from a large house price appreciation.



owner savers, who enjoy from their increased housing wealth; see Figures 7 and 8. This corresponds to the evidence of Mian, Rao, and Sufi (2013), who document large consumption responses out of housing wealth, and the theoretical analysis of Berger et al. (2017), which I augment in Proposition 4.3. Savers' consumption increases approximately 25% during the boom, constituting to a large consumption elasticity close to 1 out of increasing housing wealth: savers' net worth also increases approximately 25% during the housing boom, and the increased consumption is thus due to a wealth effect. This additional consumption is partially produced by the additional production due to improved capital allocation, as entrepreneurs hold more of the productive capital stock after the shock. The entrepreneurs are, however, not driving the consumption boom, since their net worth is only increased by approximately 5% at the impact of the shock.

Moreover, while the entrepreneurs' net worth increases slightly at the impact of the shock, the effect is not long-lasting due to their relatively high consumption propensity. While the heterogeneity in the agents' productivity is crucial for my mechanism, the house price effect is produced by a shift in the savers' portfolio incentives. Entrepreneurs enjoy from their higher productivity, but the boom mostly benefits the savers, who own the houses.

6 Empirical tests

This paper proposes a new mechanism for explaining how credit relaxations may disproportionately affect house prices. Relaxing a collateral credit constraint leads to increased aggregate productivity, which generates a positive house price effect via the general equilibrium demand channel. Therefore, a key testable prediction of the model is whether increased economic activity indeed generates a causal impact on house prices and, in particular, how strong that channel is quantitatively.

Existing literature suggests substantial income elasticity of housing demand (Davis and Ortalo-Magné, 2011). Therefore, we should expect economic growth to generate a large house price effect, given a fixed housing supply in the short run. In order to further quantify the combined effect of a collateral credit relaxation on economic growth and the effect of growth on house prices, I use County Business Patterns data on employment together with Zillow house price data, to estimate this impact. I exclude the real estate and residential construction sectors from the analysis to make sure none of the results are driven by the specifics of the housing boom-bust cycles during my study period of $2000 - 2022$ (Kerr, Kerr, and Nanda, 2022). I also restrict the employment and establishment growth rates between 10th and 90th percentiles to remove outliers and observations with infinite values. For a more thorough description of the data, I refer to Appendix B.

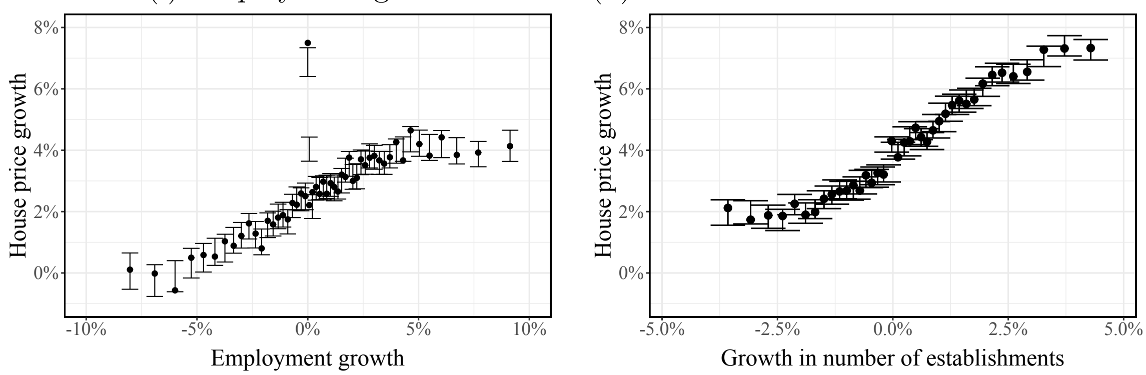
My analysis consists of two components. First, in Appendix B.2, I reproduce the results of Adelino, Schoar, and Severino (2015) to produce an estimate of how house prices affect economic activity, measured by employment. Second, in section 6.1, I use the Bartik instrument to estimate the elasticity of how an increase in employment affects house prices. I conclude in section 6.2 by combining these estimates to evaluate the quantitative significance of the feedback mechanism. In all these empirical analyses, I interpret the business capital of my model to correspond to land in the data and use house prices as a proxy for land valuations.

I note that the empirical contribution of this analysis relative to the existing literature is modest and this analysis is only provided to produce additional back-of-

Figure 9: Business activity and house prices

The figure plots the Cattaneo et al. (2024) binscatter between annual growth in business activity (employment, number of establishments) and house price growth across U.S. counties during 2000 – 2022. House price data is from Zillow. Employment and establishments counts are from County Business Patterns data and exclude residential construction and real estate sectors.

(i) Employment growth (ii) Growth in number of establishments



the-envelope evidence on the quantitative significance of my mechanism.[27]

6.1 Business activity and house prices in the data

In Figure 9, I plot the Cattaneo et al. (2024) binscatters between two measures of business activity, annual county-level employment growth and growth in number of establishments, respectively, and house price growth rates. Both measures of business activity demonstrate a strong positive correlation with house price growth rates during 2000 – 2022.

I further test this relationship in a county-level annual panel regression of the Zillow house price index on employment growth, instrumented with the Bartik (1991) instrument, across U.S. counties during 2000 – 2022:

$$\text{House price growth}_{it} = \theta_i + \delta_t + \beta \, \text{Employment growth}_{it} + \varepsilon_{it},$$

where θ_i are county and δ_t year fixed effects. I weigh the regression with the county-

[27] My modest empirical contribution here is two-fold: (i) while Adelino et al. (2015) document the effect of house prices on home equity refinancing and HELOC loans, they do not find effects on commercial and industrial lending in bank Call Reports data. I use the Community Reinvestment Act business loan data from 2000 – 2022 to augment their analysis and document similar elasticities for business loan originations than Adelino et al. (2015) find for home equity loans in their analysis; (ii) I combine these effects of house price growth rates on employment with the reverse effect of employment on house prices using the Bartik instrument. This combination allows for producing a quantitative evaluation of the combined feedback mechanism. This augments earlier work by Liebersohn (2017), who considers the effect of "Bartik shocks" on house prices.

Table 2: Employment growth and house prices

This table documents the results for a county-level regression of annual house price growth rates on county-level employment growth during 2000 – 2022. The growth rate in employment is instrumented with the Bartik instrument. The regressions are weighted by county-level (log) population in 2010. The employment growth rate is restricted between 10th and 90th percentiles.

	Dependent Variable:			
	Employm. growth	House price growth		
	First stage	*Reduced form*	*OLS*	*IV (2SLS)*
	(1)	(2)	(3)	(4)
Employm. Bartik	0.567***	0.344***		
	(0.075)	(0.074)		
Employm. growth			0.127***	0.606***
			(0.010)	(0.100)
County FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	26,978	26,978	26,980	26,978
R^2	0.243	0.505	0.508	

Clustered (county) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

level (log) population in 2010.

For the IV regression, I define the Bartik instrument in a standard manner, interacting the aggregate industry growth rates with county-level industry composition, both measured annually:

$$\text{Bartik}_{it} \equiv \sum_{l=1}^{N} w_{il,t-1} g_{lt},$$

where industries (l) are 2-digit NAICS industries (excluding real estate and residential construction), g_{lt} is the nationwide employment growth rate in industry l between years $t-1$ and t, and $w_{il,t-1}$ is the employment share of industry l in county i in year $t-1$. The first stage regression now is given by

$$\text{Employment growth}_{it} = \alpha_i + \lambda_t + \rho \, \text{Bartik}_{it} + \varepsilon_{it},$$

where α_i and λ_t are county and year fixed effects, respectively.

I report the results in Table 2 and, respectively, for the analogous regression using

Table 3: Establishment growth and house prices

This table documents the results for a county-level regression of annual house price growth rates on growth in the number of establishments during 2000 – 2022. The growth rate in the number of establishments is instrumented with the Bartik instrument. The regressions are weighted by county-level (log) population in 2010. The establishment growth rate is restricted between 10th and 90th percentiles.

	Dependent Variable:			
	Estab. growth	House price growth		
	First stage	*Reduced form*	*OLS*	*IV (2SLS)*
	(1)	(2)	(3)	(4)
Estab. growth Bartik	0.547^{***}	0.240^{**}		
	(0.047)	(0.118)		
Estab. growth			0.283^{***}	0.438^{**}
			(0.017)	(0.212)
County FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	39,556	39,556	39,556	39,556
R^2	0.321	0.555	0.560	

Clustered (county) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

establishment growth rates in Table 3. The instrumented elasticities are statistically highly significant and economically substantial, 0.606 for employment and 0.438 for establishment growth rates, respectively, for both measures of business activity.

The instrumented elasticity for employment is substantially higher than the OLS estimate. This is due to two reasons. First, for privacy reasons, the County Business Patterns data only provides "employment flags" showing the size class of the county-industry pair for such pairs, where there are not enough firms to maintain the anonymity of the data. In my analysis, which uses industry-level data to exclude the real estate and residential construction sectors, I encode these employment flags using the midpoint of the specified employment class. For details, I refer to section B.1.2, but note here that this does bring measurement error into the analysis, generating attenuation bias in the OLS estimate.

Second, the OLS estimate likely reflects the fact that the Bartik IV captures the local average treatment effect (LATE) among the complier counties that respond to the Bartik shock. For example, if the aggregate industry growth rate is driven by a demand shock in a particular 4-digit industry, it is plausible that other 4-digit industries within the same 2-digit industry would not respond to such a shock. Therefore, the industry exposure to the particular 2-digit industry is an imperfect measure of the county's exposure to such a shock, and not all counties with a high industry share in that 2-digit industry would respond to the shock similarly. The IV specification, therefore, only captures the local average treatment effect among those counties that, in reality, have true exposure to the particular 4-digit industry experiencing the shock.

Consequently, one should be cautious about how to interpret the results of this IV specification. In the following subsection, I use these estimates to produce a quantitative evaluation of the potential significance of my model mechanism. It should, however, be noted that this analysis is merely suggestive, and further empirical research testing my model mechanism is highly justified. In particular, the underlying assumption in using the LATE estimates in this analysis is that the economic shocks driving the house price fluctuations are broad-based enough to generate aggregate responses throughout the economy instead of merely affecting a small subset of complier industries.

6.2 Aggregation

The empirical analyses in sections B.2 and 6 aim to gauge whether the quantitatively substantial house price effects observed in section 5 are broadly consistent with the data. To make progress on this question, I provide a back-of-the-envelope calculation building on the reduced form estimates in Tables 2, 5, and 7.

I consider a collateral credit relaxation to reflect the relaxation of credit conditions in the early 2000s preceding the financial crisis. Indeed, Duca et al. (2011) document that the loan-to-value (LTV) ratios for mortgage originations increased from approximately 85% in 2000 to 95% in 2005. These LTV ratios, however, incorporate considerable heterogeneity. Levitin and Wachter (2011) document that a substantial fraction of mortgages originated at LTV ratios exceeding 100%, whereas An and Sanders (2010) show that for commercial mortgages, in particular, LTV ratios increased from the average of 73% pre-crisis to 82% during the peak of the boom. In particular, the exact calibration of the credit shock is complex without detailed microdata on the LTV ratios for the particular class of entrepreneurs who are the most relevant for my mechanism. Therefore, and for comparability with the existing literature, I follow Greenwald (2018) and consider a credit relaxation, where the LTV constraint is relaxed from 85% to 99%, i.e., a 16.5% increase in real estate collateral when assuming that loans are originated at the credit constraint (or at the relevant pricing discontinuity).

For the sake of argument, I assume that the existing market equilibrium is not supported after such an exogenous credit relaxation; I look for a new equilibrium where house prices have increased by x %. Given the shock, the total housing collateral has then increased by a factor of $1.165 \cdot (1 + x/100)$.

In Appendix B.2.1—cf. equation (B.2)—I consider the effect of house price shocks on business loan originations. I interpret the results of this analysis as describing how real estate collateral affects loan originations. I estimate an instrumented elasticity of 1.91, implying that the credit shock should (in this back-of-the-envelope calculation) imply a $1.91 \cdot 100 \cdot [1.165 \cdot (1+x/100) - 1]$ percent increase in business loan originations. By column (5) of Table 7 this translates to $0.480 \cdot 1.91 \cdot 100 \cdot [1.165 \cdot (1 + x/100) - 1]$ increase in employment, which by column (4) of Table 2 implies $0.606 \cdot 0.480 \cdot 1.91 \cdot 100 \cdot [1.165 \cdot (1 + x/100) - 1]$ increase in house prices. Assuming that land, the price of which is proxied by house prices, is the relevant collateral for the entrepreneur, this must be consistent with the original assumption of a x % increase in house prices:

$$0.606 \cdot 0.480 \cdot 1.91 \cdot 100 \cdot [1.165 \cdot (1 + x/100) - 1] = x.$$

This linear equation is solved by $x = 25.9\%$, explaining 64.8% of the total 40.0% real house price increase preceding the financial crisis during Q1/2003 – Q1/2007.

While this back-of-the-envelope analysis is somewhat sensitive to the elasticities used in the calculation and is not robust for the Lucas critique, it demonstrates that the estimated elasticities relevant to my house price mechanism are large enough to plausibly generate substantial house price effects.[28] This augments the quantitative evidence produced in section 5. These combined quantitative and reduced form empirical evidence suggest that the mechanism proposed in this paper may be useful for understanding how credit conditions affect house prices.

7 Conclusion

This paper presents a new theoretical framework for understanding the role of credit supply shocks in driving significant fluctuations in house prices, a phenomenon often referred to as the ”house price volatility puzzle.” While much of the existing literature focuses on household demand, this study emphasizes the impact of credit-constrained entrepreneurs who allocate capital between business investments and residential real estate. By incorporating these entrepreneurial dynamics, the model offers a fresh perspective on how capital allocation decisions affect asset prices, particularly in the housing market, thereby offering a meaningful extension to current theories of credit-driven house price booms.

The proposed mechanism suggests that by increasing entrepreneurs' access to loans, credit relaxation sets off a chain of reallocation effects that drive up house prices through two main channels. First, the model shows that increased capital demand from productive entrepreneurs leads less productive investors to rebalance their portfolios towards housing, thus impacting house prices. Second, the resulting higher productivity and aggregate income drive up demand for housing services, creating

[28] For example, using the elasticities from the regressions with a full set of control variables (1.74 and 0.379 respectively for loan originations and employment growth), the house price increase associated with the above credit relaxation would amount to 12.3%, i.e., 30.8% of the total 40.0% real house price increase. Note that while estimating the price effects using nominal house price data, when estimated in log differences, price inflation is captured by the regression fixed effects, and the estimated elasticities govern real price effects.

a sustained rent increase. This nuanced view underscores the importance of portfolio rebalancing and the role of housing as a "neutral asset" toward which investors rebalance their portfolios when facing capital demand shocks.

My quantitative analysis aligns with observed patterns during the U.S. housing boom of the early 2000s. The model demonstrates a 52.9% rise in house prices, closely mirroring the growth observed in this period. Furthermore, the model's inclusion of frictionless rental markets and calibrated housing supply elasticity suggests robustness against some of the main critiques in the literature.

Ultimately, this study contributes to the literature by providing a mechanism that not only explains rapid increases in house prices but also captures the dynamics of the ensuing bust without a shock reversal. The findings have significant implications for policymakers, highlighting the importance of firms' credit access to asset and housing markets, with implications for aggregate productivity and welfare.

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A House price amplification in a one-period model

In this section I present a simplified static model, which can be used to further illustrate some of the model mechanics.

A.1 The model environment

The one-period economy has a unit mass of two types of agents: entrepreneurs and savers. All agents are price-takers acting in a competitive economy. In the morning of the period, the agents wake up with some pre-existing portfolio holdings of productive capital k, housing h, and bonds b. Capital and housing are in fixed supply K and H, respectively. Bonds are in zero net supply $b + \underline{b} = 0$, where b denotes the combined holdings of the entrepreneurs and $\underline{b}$ the combined holdings of the savers.

For simplicity, I assume that the initial portfolios are identical within a type. I look for a symmetric equilibrium, where agents of the same type make identical choices, and the economy aggregates to two representative agents, one for each type. In what follows, I consider the problem of these representative agents. I denote the saver quantities with underlined objects when emphasizing their different behavior from the entrepreneurs.

During the morning, the agents freely trade on the assets. Capital and housing prices are endogenous and denoted by p_k and p_h, respectively. Since there is no consumption during the morning, when asset trades take place, there is no real interest rate. I normalize the nominal interest rate to zero, implying that the bond is a zero-return savings technology from the morning until the afternoon. The trading is subject to short-selling constraints $k, h \geq 0$ and to a collateral constraint

$$-b \leq (1 - \phi_k)p_k k + (1 - \phi_h)p_h h. \tag{A.1}$$

In the afternoon, production occurs, possible debt positions are cleared, and consumption takes place in that order. All agents derive utility from non-durable consumption c and from housing services s with utility $u(c^\alpha s^{1-\alpha})$, where $u(\cdot)$ satisfies Inada conditions. The consumption good c, which is the numéraire in the economy, is produced with $ak^\gamma, \gamma \in (0, 1]$, technology, where the saver's and the entrepreneur's productivity parameters are given, respectively, by $\underline{a}$ and a. I assume that the entrepreneur is more productive at any given level of capital holdings: $a > \underline{a}$. Housing services are produced one-to-one by houses h and are traded in the afternoon at a frictionless rental market with an endogenous rental price r_s.

For an initial portfolio (k_{io}, h_{io}, b_{io}), an agent i solves

$$\begin{aligned} \max_{c_i,s_i,k_i,h_i,b_i} \quad & u(c_i^{\alpha} s_i^{1-\alpha}) \\ \text{s.t.} \quad c_i + r_s s_i &\leq a_i k_i^{\gamma} + r_s h_i + b_i. \\ p_k k_i + p_h h_i + b_i &= p_k k_{io} + p_h h_{io} + b_{io}, \end{aligned} \tag{A.2}$$

subject to the collateral constraint (A.1) and the short-selling constraints on h and k.

The next proposition replicates Proposition 3.1 in section 2.

Proposition A.1 (Irrelevance of housing collateral)**.** *A financially constrained agent will never own housing, i.e., $h = 0$ necessarily. Consequently, relaxation of the housing collateral constraint ϕ_h does not affect house prices.*

A.2 Equilibrium prices

I denote the (representative) entrepreneur's ownership share of capital by $\psi_k \equiv k/K$, the average productivity of capital by

$$A(\psi_k) \equiv \frac{ak^{\gamma} + \underline{a}\,\underline{k}^{\gamma}}{K} = \left[a\psi_k^{\gamma} + \underline{a}(1-\psi_k)^{\gamma}\right] K^{\gamma-1},$$

the total wealth in the economy by $N \equiv p_k K + p_h H$, aggregate consumption by $C \equiv c + \underline{c}$, and the value of total capital stock relative to the total wealth in the economy by $\nu \equiv p_k K/(p_k K + p_h H)$. The following result characterizes the endogenous prices of the model in terms of ψ_k, ν, and the aggregate consumption propensity C/N and replicates Theorem 3.2 in section 3.

Theorem A.2. *The rental rate r_s, house price p_h, and price of capital p_k can be characterized via the following formulas:*

$$r_s = A(\psi_k)\frac{K}{H}\frac{1-\alpha}{\alpha} \quad \textit{and} \quad p_h = A(\psi_k)\frac{K}{H}\frac{(1-\nu)}{C/N},$$

as well as

$$p_k = A(\psi_k)\frac{\nu}{C/N}.$$

A.3 Elastic capital supply by savers: $y = ak, \gamma = 1$.

I continue the model analysis by first concentrating on the case of linear ak production technology, i.e., assuming $\gamma = 1$. I will show that, in this case, the supply of capital by savers to financially constrained entrepreneurs is fully elastic, implying that the collateral capital price is not affected by the credit relaxation. This has the advantage that, without such a price effect on the collateral capital, the model has no financial accelerator. Therefore, the $\gamma = 1$ setup provides a clean presentation of the house price mechanism, demonstrating that the house price effect is entirely independent of the existing financial accelerator mechanisms (Kiyotaki and Moore, 1997; Iacoviello, 2005; Liu et al., 2013).

I begin by characterizing the model equilibrium, assuming the collateral constraint is not binding for either (representative) agent. This will provide a benchmark against which to contrast the results of the next subsection, which will consider constrained agents.

Proposition A.3. *Suppose $\gamma = 1$ and that the collateral constraint is not binding for either agent. Then*

$$\frac{C}{N} = \alpha \quad \textit{and} \quad \nu = \alpha.$$

When the collateral constraint is not binding, the portfolio choices are unrestricted, and the share of non-durable consumption out of total wealth is just given by its preference share.

Similarly, the relative valuation of assets, described by ν, is also entirely determined by preferences for non-durable consumption vs. housing, as $\nu = \alpha$.

Proposition A.4. *Suppose $\gamma = 1$ and that the collateral constraint is not binding. Then $\psi_k = 1$.*

For $\gamma = 1$, the unconstrained economy is therefore quite simple with $p_k = a$ and $r_s = p_h = \frac{aK}{H}\frac{1-\alpha}{\alpha}$. We will use this as an intuitive benchmark when considering the effects of the collateral constraint.

A.3.1 Collateral constraint

Concentrating on the case of a constrained entrepreneur, I make the following assumption.

Assumption A.5. Suppose the entrepreneur initial portfolio (k_o, h_o, b_o) satisfies

$$\frac{k_o}{K} + \frac{a}{\underline{a}}\frac{1-\alpha}{\alpha}\frac{h_o}{H} + \frac{b_o}{\underline{a}K} < \phi_k. \tag{A.3}$$

The main content of this assumption is that the entrepreneur cannot be wealthy enough not to be financially constrained:

Lemma A.6. *Under assumption A.5 and with $\gamma = 1$, the collateral constraint is binding for the entrepreneur.*

This result resembles that of Evans and Jovanovic (1989), who show in a similar model that the set of financially constrained agents can be characterized in terms of their productivity and assets, which are also two main ingredients of equation (A.3). In my model, the value of an agent's portfolio depends on endogenous asset prices, which complicates the characterization of which parameter combinations lead to constrained agents, compared to assuming exogenous assets. Equation (A.3), however, gives a sufficient condition for a binding collateral constraint in the spirit of Evans and Jovanovic (1989, Figure 1).

Next, let us consider the consumption propensities of the agents. We have the following result.

Proposition A.7. *Suppose $\gamma = 1$ and that the collateral constraint is binding for the entrepreneur. Then*

$$\frac{\underline{c}}{\underline{n}} = \alpha \quad \textit{and} \quad \frac{c}{n} = \alpha\left[1 + \underbrace{\frac{a - \underline{a}}{\phi_k p_k}}_{\substack{\textit{Excess return}\\ \textit{on capital}}}\right].$$

In case of a binding collateral constraint, the entrepreneur's ownership of productive capital is restricted. In this case, the entrepreneur has an absolute advantage in production, which gives them excess return on their capital investment over the risk-free interest rate (normalized to 0). This excess return justifies higher consumption propensity out of their net worth for the entrepreneurs than for the savers. From the point of view of the entrepreneurs, the productive capital – which effectively determines their net worth – is underpriced, and they would prefer to purchase more of it. The collateral constraint, however, prevents them from financing additional capital purchases.

A.3.2 House price amplification

The following Proposition describes how the collateral constraint affects relative prices, described by ν.

Proposition A.8. *Under Assumption A.5 and with $\gamma = 1$, the capital share of total wealth ν can be characterized in terms of the capital allocation ψ_k as*

$$\frac{\alpha}{\nu} = \alpha + (1-\alpha)\frac{A(\psi_k)}{\underline{a}}. \tag{A.4}$$

In particular, $\nu = \nu(\psi_k)$ is a decreasing function of ψ_k with $\nu \le \nu(0) = \alpha$.

To unpack the contents of this result, recall first that the capital allocation ψ_k can be considered the state variable of the model when the entrepreneurs are credit-constrained. The initial portfolio endowments of the entrepreneur and the saver determine their relative wealth positions, which induce some capital allocation described by ψ_k after trading on the assets. More precisely, by Proposition A.1 the wealth of a credit-constrained entrepreneur is given by

$$n = p_k k + b = p_k k - (1-\phi_k)p_k k = \phi_k p_k k,$$

which yields

$$k = \psi_k K = \frac{n}{p_k \phi_k} = \frac{n}{\underline{a}\phi_k}.$$

Noting that the entrepreneur's wealth is determined by their initial portfolio, the above result thus describes the relative price of capital vs. housing in terms of the state of the economy characterized by capital allocation ψ_k, which in turn is determined by the collateral constraint parameter ϕ_k. A credit relaxation, i.e., a decrease in ϕ_k, generates reallocation of capital, whereby entrepreneurs increase their capital holdings.

To discuss, how this affects asset prices, recall first that $A(\psi_k) \equiv a\psi_k + \underline{a}(1-\psi_k)$ is the average productivity in the economy. Proposition A.8 states that when more capital is in the hands of the entrepreneur, i.e., economywide aggregate productivity is higher, the relative price of housing is higher. This result is a straightforward implication of the saver's portfolio allocation problem described by their first-order conditions:

$$\underbrace{\frac{\underline{a}}{p_k}}_{\text{Production yield}} = \underbrace{\frac{r_s(\psi_k)\uparrow}{p_h\uparrow}}_{\text{Rental yield}}. \tag{A.5}$$

By Theorem A.2, the rental price r_s varies only with the aggregate productivity $A(\psi_k)$. On the other hand, since the savers also invest in the zero-return bond—thus financing the entrepreneurs' capital purchases—it must be the case that $p_k = \underline{a}$ whenever the entrepreneurs are credit-constrained. The capital allocation, therefore, does not affect the capital price at all, but due to the general equilibrium demand channel, it matters for rental rates.

Indeed, by Theorem A.2, we know that the rental rate r_s increases with ψ_k. The improved capital allocation increases average productivity, which increases aggregate income and, therefore, aggregate demand that will affect the rents. In order to balance the no-arbitrage condition (A.5), house prices must increase. This implies that ν must decrease in ψ_k, providing the intuition for Proposition A.8.

Another important implication of $p_k = \underline{a}$ is that the price of capital does not respond to the capital allocation in the economy as long as the entrepreneurs are credit-constrained. This differs from the setup of Kiyotaki and Moore (1997), as the standard financial accelerator mechanism is completely shut down in this economy with $\gamma = 1$: a slight credit relaxation, which does not completely lift the entrepreneurs off their credit constraints does not affect the price of the collateral and does not generate a financial accelerator. Nevertheless, the relative prices of capital and housing are affected. This emphasizes that the relative price effect is entirely independent of the classic Kiyotaki-Moore financial accelerator mechanism.

I illustrate these mechanics in Figure 1. In panel (a) of the figure, savers provide the entrepreneurs with a fully elastic supply of capital at a valuation $p_k = \underline{a}$, all the way up to the total capital stock K. On the other hand, the entrepreneurs have a fully elastic demand for capital at a higher valuation $p_k = a > \underline{a}$. However, the collateral credit constraint limits the entrepreneurs' capital holdings, which in panel (a) of the figure, is represented by the vertical credit limit line. This is the most amount of capital the entrepreneurs can own, given the LTV constraint limiting their borrowing.

A relaxation of the credit constraint moves this line horizontally, allowing the entrepreneurs to purchase more of the capital stock. Due to the fully elastic capital supply by the savers, the price of capital, however, does not respond to the credit shock. Instead, we observe the price effect on the housing market. The credit relaxation induces a capital reallocation, which increases aggregate output in the economy affecting rents via the general equilibrium, thus increasing house prices.

Much housing boom-bust literature has concentrated on increased housing demand due to a credit supply shock. The mechanism here is, however, different. While it manifests as a housing demand shock (via the general equilibrium), it is not driven by increased demand for housing services *per se*. Instead, the combination of increased productivity and the portfolio reallocation incentives of certain investors produce a disproportionate increase in house prices.

Indeed, the broad implication of this result is that in addition to the well-known financial accelerator mechanisms, the collateral credit constraint can generate changes in the relative prices of real assets when asset valuations and asset demand vary between mutually heterogeneous investors. The mechanism works through two different channels: (i) the portfolio rebalancing channel as less productive investors reallocate their portfolios toward housing, in order to equalize investment yields, and (ii) the general equilibrium demand channel as housing rents increase with improved capital allocation that increases aggregate productivity. In a competitive economy with perfectly elastic capital supply by the savers (such as here), the price of capital does not respond. However, the result is more general, as I will demonstrate in the following section.

A.4 Inelastic capital supply: $y = ak^{\gamma}, \gamma < 1$.

The analysis in section A.3.2 demonstrates that the key equation determining the extent of house price amplification is that of the saver's no-arbitrage condition, which with $\gamma < 1$ takes the form

$$\frac{\gamma \underline{a}\,\underline{k}^{\gamma-1}}{p_k} = \frac{r_s}{p_h}. \tag{A.6}$$

The numerator on the left-hand side of the equation includes the marginal productivity of saver's capital, which in case of a linear technology was directly given by the saver's productivity parameter $\underline{a}$, but more generally takes the form $\gamma \underline{a}\,\underline{k}^{\gamma-1} = \gamma \underline{a}\,\underline{(}1-\psi_k)^{\gamma-1}K^{\gamma-1}$. A credit relaxation affecting capital allocation also affects this marginal productivity, with two main implications for asset prices.

First, part of the credit shock is now absorbed by capital price p_k, as the supply of capital from savers to entrepreneurs is no longer fully elastic. Increasing entrepreneur capital share ψ_k also increases the saver's marginal capital productivity. If this direct effect on capital productivity is more significant than the indirect, general equilibrium effect on the rental rate r_s, house prices might not respond as strongly as the price of capital.

Second, since the price of capital is now responding to a credit shock, this will introduce a financial accelerator à la Kiyotaki and Moore (1997) into the model.[29] Credit relaxation increases the price of capital, further relaxing the credit constraint and generating a strategic complementarity between asset prices and credit access. The exact magnitudes of these forces depend on the model parameters. However, in-

[29]Recall that by Proposition A.1, in equilibrium, credit-constrained entrepreneurs only own business capital and not housing.

dependent of these parameters, it is necessarily true that for sufficiently constrained agents (i.e., sufficiently low entrepreneur capital share ψ_k), the price of housing increases disproportionately for a small credit shock:

Theorem A.9. *Let*

$$g(\psi_k) \equiv \frac{p_h}{p_k}.$$

Then $g'(0) > 0$.

By (A.6), we can write the ratio of house and capital prices as a function of ψ_k as[30]

$$g(\psi_k) = \frac{p_h}{p_k} = \frac{r_s}{\gamma \underline{a}\, \underline{k}^{\gamma-1}} = \frac{A(\psi_k)}{\gamma \underline{a}(1-\psi_k)^{\gamma-1} K^{\gamma-1}} \frac{K}{H} \frac{1-\alpha}{\alpha}. \tag{A.7}$$

In particular, the price ratio depends on the credit constraint parameter ϕ_k only through its effect on the capital allocation ψ_k. As long as the entrepreneur cannot afford to buy too much of the capital, i.e. ψ_k is close to zero, a (small) relaxation of the constraint allowing an increase in ψ_k will disproportionately increase the price of housing (i.e., $g' > 0$).

Since Theorem A.7 already qualitatively establishes the price amplification mechanism discussed in detail for the case $\gamma = 1$, I omit the full details of the equilibrium characterization for $\gamma < 1$ and, instead, proceed to discuss the financial accelerator, which embeds a new force due to a novel interaction between the capital and housing markets.

A.4.1 Financial accelerator

Recall that by Proposition A.1, the wealth of a credit-constrained entrepreneur is given by

$$n = \phi_k p_k k.$$

By the discussion in section A.3, in the benchmark case of $\gamma = 1$, we have $p_k = \underline{a}$. For a given level n of entrepreneur wealth, this implies that in any model equilibrium,

[30] Note that if the entrepreneur is not credit-constrained, then combining the no-arbitrage conditions for both agents yields

$$\frac{a}{\underline{a}} = \left(\frac{\psi_k}{1-\psi_k}\right)^{1-\gamma},$$

which pins down the capital allocation ψ_k, in which case the state of the economy is characterized by the allocation of housing $\psi_h = h/H$. Therefore, we can consider ψ_k as a state variable describing the state of the economy only in the case of credit-constrained entrepreneurs (when $\psi_h = 0$ necessarily). In particular, the fact that the right hand side of (A.7) equals zero at $\psi_k = 1$ for $\gamma < 1$ has no economic meaning, as $\psi_k = 1$ can never represent an equilibrium of the model for $\gamma < 1$.

the entrepreneur capital allocation $k = \psi_k K$ is characterized by

$$n = \phi_k \underline{a} \psi_k K,$$

which yields

$$\psi_k = \frac{n}{\phi_k \underline{a} K}, \quad \gamma = 1. \tag{A.8}$$

In the case of perfectly elastic capital supply by the savers, as discussed in section A.3 and depicted in Figure 1, the credit constraint does not affect the price of capital. Therefore, a credit relaxation does not affect the wealth of a credit-constrained entrepreneur, who is fully invested in the capital stock. Holding wealth constant before and after a credit relaxation, $n = n_o = n_1$, this implies that the capital allocation before and after a reduction in $\phi_k = \phi_{kko} \rightarrow \phi_{kk1} < \phi_{kko}$ must by equation (A.8) satisfy

$$\frac{\psi_{k1}}{\psi_{ko}} = \frac{\phi_{kko}}{\phi_{kk1}}, \tag{A.9}$$

for $\gamma = 1$. In particular, the credit constraint parameter ϕ_k completely characterizes the capital allocation.

For $\gamma < 1$, the price of capital does not remain constant given a credit relaxation. I consider an entrepreneur with initial wealth $n_o = \phi_{kko} p_{ko} k_o = \phi_{kko} p_{ko} \psi_{ko} K$, and study, how the capital allocation is affected by a sudden decrease $\phi_{kko} \rightarrow \phi_{kk1}$, when taking into account that the change in the capital allocation will, for $\gamma < 1$, also affect the price of capital and, therefore, the entrepreneur wealth. We have the following theorem.

Theorem A.10. *Let $n_o = p_{ko} k_o + b_o$ and ϕ_{kko} be such that the entrepreneur is credit-constrained in the equilibrium. Consider such a reduction in $\phi_{kk} = \phi_{kko} \rightarrow \phi_{kk1}$ that the entrepreneur remains credit constrained in the new equilibrium, with the entrepreneur's initial portfolio given by k_o and b_o. Then, the new equilibrium allocation ψ_{k1} satisfies*

$$\frac{\psi_{k1}}{\psi_{ko}} = \frac{\phi_{kko}}{\phi_{kk1}} + \underbrace{\frac{1 - \phi_{kko}}{\phi_{kk1}} \left[1 - \left(\frac{1 - \psi_{k1}}{1 - \psi_{ko}} \right)^{1-\gamma} \right]}_{\textit{Financial accelerator}}. \tag{A.10}$$

Note first that when $\gamma = 1$, Theorem A.10 merely replicates (A.9), i.e., the case when capital price does not respond to the credit shock and the financial accelerator is shut down. In the case of $\gamma < 1$, however, the right hand side of (A.10) now includes an additional term, reflecting the financial accelerator. A collateral credit relaxation reducing the value of ϕ_k allows the entrepreneurs to purchase more of the business capital. Due to the financial accelerator, this improvement in the capital allocation further increases the right hand side of equation (A.10), which then implies a more substantial increase in the value of ψ_k than what would be the case without the accelerator.

A.4.2 Strategic complementarity and multiple equilibria

While the idea behind the financial accelerator discussed in the previous section is well-known (Kiyotaki and Moore, 1997), when combined with the housing market, the interaction between the housing and capital markets provides a new and additional source of amplification. This force is due to the disproportionate increase in house prices discussed in detail in section A.3 and is strong enough to generate multiple equilibria for the model.[31]

Theorem A.11. *Suppose $\gamma < 1$. Then there exists an entrepreneur initial portfolio (k_o, h_o, b_o) with $k_o = 0$ such that the model admits two possible equilibria (after trading).*

Intuitively, when a productive entrepreneur initially only holds housing, the equilibrium house price amplification mechanism discussed in sections A.3 and A.4 producing a disproportionate house price boom now disproportionately benefits the entrepreneur. This amplifies the financial accelerator mechanism enough to potentially produce multiple equilibria.

Note that the intuition behind the proof of this result is also simple. Assuming an entrepreneur portfolio with $k_o = 0$, their wealth is given by

$$n = p_h h_o + b_o = A(\psi_k) K \frac{h_o}{H} \frac{1-\alpha}{\alpha} + b_o, \tag{A.11}$$

which depends on the endogenous capital allocation ψ_k. On the other hand, we know that, if the entrepreneur is credit-constrained after trading, their wealth must equal

$$\begin{aligned} n = p_k k + b &= p_k k + (1-\phi_k) p_k k = \phi_k p_k k \\ &= \phi_k \gamma \underline{a}\, \underline{k}^{\gamma-1} k \\ &= \phi_k \gamma \underline{a} (1-\psi_k)^{\gamma-1} \psi_k K^{\gamma}. \end{aligned} \tag{A.12}$$

The question is, under which conditions does this nonlinear system of equations, (A.11)

[31] Theorem A.11 provides a sufficient, but not a necessary, condition for the existence of multiple equilibria.

and (A.12), have two solutions. Note that this is a system of two equations on two unknowns n and ψ_k.

To illustrate the mechanism, I first calibrate the model parameters to produce a quantitative example. I begin by normalizing $\underline{a} = 1, K = 1$ and $H = 1$. For calibrating the heterogeneity in productivity, I rely on Fagereng, Guiso, Malacrino, and Pistaferri (2020, Table 7), who report that among business owners, the return to net worth is 6.28% in the top quartile and 10.09% in the top decile of the return distribution. I use the ratio between these two to calibrate $a = 1.61$. Furthermore, I set $\gamma = 0.5$ and $\alpha = 0.8$, the latter of which reflects an approximate 20% consumption share of housing net of utilities (Davis and Ortalo-Magné, 2011).[32] I also assume the entrepreneurs initially own $h_o = 15\%$ of the housing stock with $k_o = 0$. Following the proof of Theorem A.11, I also set

$$b_o = \underline{a}K^\gamma \frac{h_o}{H}\frac{1-\alpha}{\alpha} - 0.005,$$

to guarantee that the entrepreneurs would be underwater, if the capital allocation was the worst possible with $\psi_k = 0$. Finally, I set the collateral constraint parameter at $\phi_{kko} = 0.15$ and $\phi_{kk1} = 0.01$, corresponding to an 85% loan-to-value ratio, relaxed to 1% by the credit shock (Greenwald, 2018).

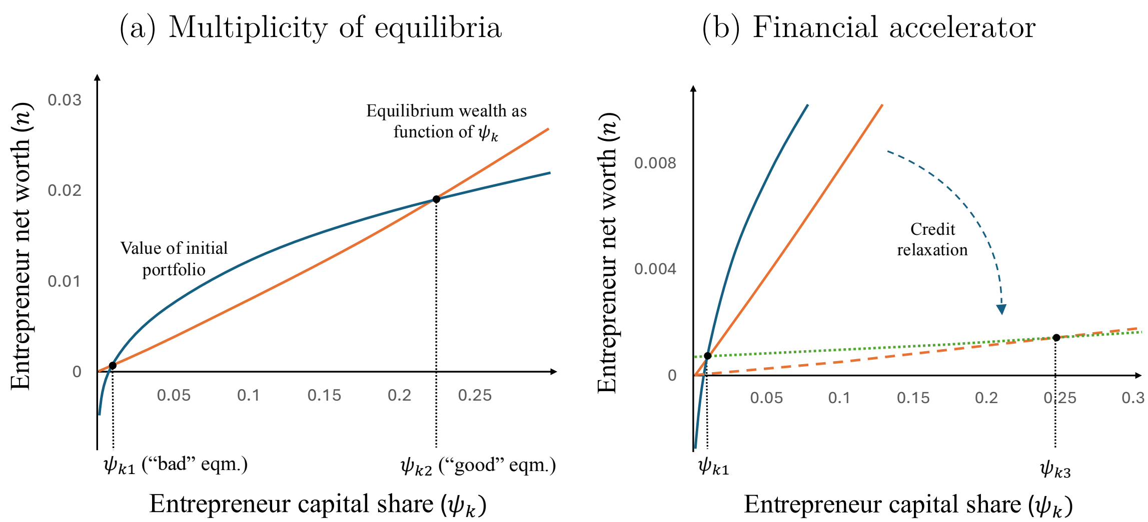
In panel (a) of Figure 10, I depict the entrepreneurs' net worth n as a function of the capital allocation ψ_k, showing the relations specified by both (A.11) and (A.12). These equations have two solutions corresponding to two different equilibria: $\psi_{k1} = 0.0096$ ("bad" equilibrium) and $\psi_{k2} = 0.225$ ("good" equilibrium). In the good equilibrium, the levered entrepreneurs trade off their 15% housing share to 22.5% ownership share on the productive capital. However, if everyone believes that the output and prices should be lower, the economic fundamentals also support another equilibrium with much worse capital allocation, with the entrepreneurs owning only 0.96% of the capital stock.

This multiplicity relies on the strategic complementarity between the entrepreneurs' capital holdings and house prices. Higher house prices allow the entrepreneurs, whose initial holdings consist of housing, to take more debt to purchase more of the business capital. Higher entrepreneur business capital holdings reflect better capital allocation, which increases house prices as detailed in section A.3. Such an increase in house prices further relaxes the entrepreneurs' credit constraints generating a strong

[32] Davis and Ortalo-Magné (2011) report 24% consumption share for housing. I follow Greenwald (2018) in assuming a 4% cost for utilities to come up with a 20% housing consumption share, net of utilities.

Figure 10: Multiplicity of equilibria and the financial accelerator

The figure illustrates the strategic complementarity between house prices and the capital allocation for entrepreneurs initialized with housing wealth as well as the related financial accelerator mechanism. The orange line in panel (a) depicts the equilibrium relation between capital allocation and entrepreneur wealth for credit-constrained entrepreneurs. The curved blue line, on the other hand, shows how the value of the entrepreneur's initial portfolio depends on the equilibrium capital allocation. The model equilibria are characterized by the intersections of these two lines. Panel (b) of the figure (with rescaled y-axis) considers a credit relaxation, depicted as a rotation of the orange line. The dotted green line describes how the entrepreneur's wealth depends on the capital allocation, assuming an initial portfolio associated with the "bad" equilibrium. The new equilibrium is found at the intersection of the dashed orange and dotted green lines, where the entrepreneur wealth relation and the equilibrium condition coincide.



feedback mechanism, yielding multiple equilibria. These two solutions (the "good" and "bad" equilibria) are shown in panel (a) of Figure 10

Consider now an economy starting from the bad equilibrium and facing an exogenous, unanticipated credit relaxation, reducing the collateral credit parameter ϕ_k from $\phi_{kk1} = 0.15$ to $\phi_{kk2} = 0.01$, corresponding to 99% maximal loan-to-value ratio. The economy must readjust to satisfy the new equilibrium conditions associated with this new parameter setup. I depict this adjustment in panel (b) of Figure 10.

The capital allocation adjusts according to the accelerator mechanism described by equation (A.10). Note that $n = \psi_k = 0$ is always a solution for equation (A.12). In particular, a relaxation of the credit constraint, represented by a reduction in the value of ϕ_k, can, therefore, be visually described as a clock-wise rotation of the graph describing the relation between n and ψ_k specified by equation (A.12), and shown as the dashed orange line in panel (b) of Figure 10.

Given the agents' portfolio holdings in the initial bad equilibrium, the adjustment affecting prices will also affect the wealth of all agents. The dotted green line shows how the value of the entrepreneur's portfolio depends on the capital allocation. The new equilibrium after the credit shock is characterized by the capital allocation, for which the value of the entrepreneur portfolio coincides with the equilibrium condition described by equation (A.12). The new equilibrium is, therefore, depicted at the intersection of the dotted green and dashed orange lines in panel (b) of Figure 10.

Under the new parameter values, the bad equilibrium is no longer viable, and the credit relaxation leads to a large reallocation of capital with entrepreneurs owning $\psi_{k3} = 24.8\%$ of the capital stock after the credit relaxation. The improvement in capital allocation leads to a 44.8% increase in house prices.

While I have calibrated some of the key parameters using the data as discussed above, we should nevertheless be cautious in interpreting the quantitative significance of these results. In particular, this example is mainly produced to demonstrate the potential of the amplification mechanism, but a more realistic quantification of this mechanism in a fully specified DSGE model is left for future work, although I do provide some additional reduced form evidence in section 6. Nevertheless, here I have mainly just described a simple example demonstrating the mechanism's power in generating strong enough amplification to produce multiple equilibria, with a credit shock generating a shift in the equilibrium, from a "bad" one to a "good" equilibrium.

Indeed, on the one hand, pure differences in beliefs can generate multiple equilibria with economically substantial differences in capital allocation and house prices between two possible model equilibria. On the other hand, starting from the "bad" equilibrium, a credit relaxation may, in addition to forcing a reallocation associated with the shock, make the bad equilibrium non-viable, generating a substantial house price effect when the economy reallocates to the "good" equilibrium (associated with the new parameter values).

A.5 Proofs

Proof of Theorem A.1. Let's denote the Lagrangian associated with (A.2) by

$$
\begin{aligned}
\mathcal{L} = u(c^{\alpha}s^{1-\alpha}) &+ \lambda_1[ak^{\gamma} + r_s h + b - c - r_s s] \\
&+ \lambda_2[p_k(k_o - k) + p_h(h_o - h) + b_o - b] \\
&+ \lambda_3[(1-\phi_k)p_k k + (1-\phi_h)p_h h + b] \\
&+ \lambda_4 k + \lambda_5 h.
\end{aligned}
$$

The first order conditions for (A.2) are given by

$$0 = \frac{\partial \mathcal{L}}{\partial c} = \alpha u' c^{\alpha-1} s^{1-\alpha} - \lambda_1, \tag{A.13}$$

$$0 = \frac{\partial \mathcal{L}}{\partial s} = (1-\alpha) u' c^{\alpha} s^{-\alpha} - \lambda_1 r_s, \tag{A.14}$$

$$0 = \frac{\partial \mathcal{L}}{\partial k} = \lambda_1 a \gamma k^{\gamma-1} - \lambda_2 p_k + \lambda_3 (1-\phi_k) p_k + \lambda_4, \tag{A.15}$$

$$0 = \frac{\partial \mathcal{L}}{\partial h} = \lambda_1 r_s - \lambda_2 p_h + \lambda_3 (1-\phi_h) p_h + \lambda_5, \tag{A.16}$$

$$0 = \frac{\partial \mathcal{L}}{\partial b} = \lambda_1 - \lambda_2 + \lambda_3. \tag{A.17}$$

The last two equations now yield

$$1 + \frac{\lambda_3}{\lambda_1} = \frac{\lambda_2}{\lambda_1} = \frac{r_s}{p_h} + \frac{\lambda_3}{\lambda_1}(1-\phi_h) + \frac{\lambda_5}{\lambda_1 p_h}.$$

which simplifies to

$$1 = \frac{r_s}{p_h} - \frac{\lambda_3}{\lambda_1}\phi_h + \frac{\lambda_5}{\lambda_1 p_h}. \tag{A.18}$$

Suppose there exists an agent, who is not financially constrained and who invests in housing, i.e., $\lambda_3 = \lambda_5 = 0$. This implies that $r_s/p_h = 1$. Therefore, for any agent in the economy, it is necessarily true that

$$\frac{\lambda_3}{\lambda_1}\phi_h = \frac{\lambda_5}{\lambda_1 p_h}.$$

Since $\lambda_1 > 0$ necessarily, for any $\phi_h > 0$, we must have that a binding LTV constraint (i.e. $\lambda_3 > 0$) implies $\lambda_5 > 0$, i.e., that $h = 0$.

Suppose now that the economy does not have an agent with $\lambda_3 = \lambda_5 = 0$. Since the economy necessarily have some financially unconstrained agents, this implies that there is an agent with $\lambda_3 = 0$ and $\lambda_5 > 0$. For such an agent we have

$$1 = \frac{r_s}{p_h} + \frac{\lambda_5}{\lambda_1 p_h},$$

which implies $r_s < p_h$. On the other hand, all housing stock is now owned by financially constrained agents, which implies that there is an agent with $\lambda_3 > 0$ and $\lambda_5 = 0$. By equation (A.18), for such an agent we have

$$1 + \frac{\lambda_3}{\lambda_1}\phi_h = \frac{r_s}{p_h},$$

which implies $r_s > p_h$, which produces a contradiction. Therefore, the economy must have an agent with $\lambda_3 = \lambda_5 = 0$, which implies that for all agents with $\lambda_3 > 0$, necessarily $\lambda_5 > 0$, thus finishing the proof.

$\square$

Proof of Theorem 3.2. Equations (A.13) and (A.14) yield

$$r_s s = \frac{1-\alpha}{\alpha} c.$$

Plugging this into an agent's budget constraint gives the classic Marshallian demand functions under unit-elastic substitution:

$$c^* = \alpha[ak^\gamma + r_s h + b] \quad \text{and} \quad r_s s^* = (1-\alpha)[ak^\gamma + r_s h + b]. \tag{A.19}$$

Summing the demand function for s^* over the agents, by housing and bond market clearing conditions we have

$$r_s H = r_s[s + \underline{s}] = (1-\alpha)[ak^\gamma + \underline{a}\,\underline{k}^\gamma + r_s H] = (1-\alpha)[A(\psi_k)K + r_s H],$$

which implies

$$r_s = A(\psi_k)\frac{K}{H}\frac{1-\alpha}{\alpha}. \tag{A.20}$$

I denote by $N \equiv p_k K + p_h H = n + \underline{n}$ the total wealth in the economy. By goods market clearing we now have

$$\frac{C}{N} = \frac{A(\psi_k)K}{N} = \frac{A(\psi_k)}{p_k}\frac{p_k K}{N} = \frac{A(\psi_k)}{p_k}\nu,$$

which gives

$$p_k = A(\psi_k)\frac{\nu}{C/N}.$$

Moreover, by the definition of ν, it immediately follows that

$$p_h = p_k \frac{K}{H}\frac{1-\nu}{\nu} = A(\psi_k)\frac{K}{H}\frac{1-\nu}{C/N}.$$

$\square$

Proofs of Propositions A.3 and A.4. I rewrite the first order conditions as follows

$$\text{(A.17)} \quad \Longleftrightarrow \quad 1 = \frac{\lambda_2}{\lambda_1} - \frac{\lambda_3}{\lambda_1},$$

$$(\text{A.15}) \quad \Longleftrightarrow \quad \frac{\lambda_2}{\lambda_1} = \frac{a}{p_k} + \frac{\lambda_3}{\lambda_1}(1-\phi_k) + \frac{\lambda_4}{\lambda_1 p_k},$$

and

$$(\text{A.16}) \quad \Longleftrightarrow \quad \frac{r_s}{p_h} = \frac{\lambda_2}{\lambda_1} - \frac{\lambda_5}{\lambda_1 p_h}.$$

Solving for λ_2/λ_1 from each of these, gives

$$\frac{\lambda_2}{\lambda_1} = 1 + \frac{\lambda_3}{\lambda_1} = \frac{r_s}{p_h} + \frac{\lambda_5}{\lambda_1 p_h} = \frac{a}{p_k} + \frac{\lambda_3}{\lambda_1}(1-\phi_k) + \frac{\lambda_4}{\lambda_1 p_k}.$$

Therefore

$$1 = \frac{a}{p_k} - \frac{\lambda_3}{\lambda_1}\phi_k + \frac{\lambda_4}{\lambda_1 p_k}. \tag{A.21}$$

Suppose the collateral constraint is not binding for either agent, i.e., $\lambda_3 = 0$ for both agents. Then

$$\frac{a}{p_k} + \frac{\lambda_4}{\lambda_1 p_k} = 1 = \frac{r_s}{p_h} + \frac{\lambda_5}{\lambda_1 p_h} \tag{A.22}$$

and necessarily $\underline{\lambda}_4 > 0$. This implies that $\psi_k = 1$, and finishes the proof of Proposition A.4. Moreover, for $\psi_k = 1$, we also obtain that $p_k = a = A(\psi_k)$, which by Proposition 3.2 yields $\nu = C/N$.

Now

$$\begin{aligned}
c_i &= \alpha[a_i k_i + r_s h_i + b_i] \\
&= \alpha[a_i k_i + r_s h_i + (n_i - p_k k_i - p_h h_i)] \\
&= \alpha\left[a_i k + r_s h_i + n_i - \frac{a}{p_k} p_k k_i - \frac{r_s}{p_h} p_h h_i\right] \\
&= \alpha n
\end{aligned}$$

for both agents $i = e(ntrepreneur), s(aver)$. Note that here we used the fact that for non-binding collateral constraint $a_i \neq a$ if and only if $k = 0$. This now implies $C = \alpha N$ and therefore $\nu = \alpha$, finishing the proof of Proposition A.3. □

Proof of Proposition A.7. Suppose now that the collateral constraint binds for the entrepreneur and $\lambda_3 > 0$. It still follows exactly as in the Proof of Proposition A.3 that $\underline{c} = \alpha \underline{n}$.

For the entrepreneur, we first note that by (A.21)

$$\frac{\lambda_3}{\lambda_1}\phi_k = \frac{a - \underline{a}}{p_k}$$

Therefore, for the entrepreneur we have

$$c = \alpha\Big[(a - \underline{a})k + n\Big].$$

Since

$$n = p_k k + b = \phi_k p_k k$$

it follows that

$$c = \alpha n\left[1 + \frac{a - \underline{a}}{\phi_k p_k}\right],$$

which finishes the proof.

$\square$

Proof of Lemma A.6.

Claim 1: Assumption A.3 implies that the entrepreneur capital purchases satisfy $k < K$. Suppose to the contrary. Then entrepreneur holdings satisfy $k = K$ and $p_h = \frac{aK}{H}\frac{1-\alpha}{\alpha}$. The formula for house price follows from (A.22), which yields

$$p_h = r_s = A(\psi_k)\frac{K}{H}\frac{1-\alpha}{\alpha}.$$

These imply that

$$\begin{aligned} p_k k_o + p_h h_o + b_o &\geq p_k k + p_h h + b \\ &= p_k K + p_h h + b \\ &\geq p_k K + p_h h - (1-\phi_k)p_k K - (1-\phi_h)p_h h \\ &\geq \phi_k p_k K + \phi_h p_h h \geq \phi_k p_k K, \end{aligned}$$

where the second inequality follows from the entrepreneur collateral constraint. Since not all agents can be credit constrained, equation (A.21) implies that $p_k \geq \underline{a}$ necessarily. Therefore, dividing the above inequality by $p_k K$ yields

$$\frac{k_o}{K} + \frac{a}{\underline{a}}\frac{h_o}{H}\frac{1-\alpha}{\alpha} + \frac{b_o}{\underline{a}K} \geq \phi_k.$$

This yields a contradiction with Assumption A.3. Therefore, $k < K$ necessarily.

Claim 2: $k_o < K$ implies that the entrepreneur collateral constraint is binding.

By (A.21), if $\lambda_3 = 0$ for both agents, $\underline{k}_o = 0$. Therefore, $\lambda_3 > 0$ at least for one of the representative agents. Suppose this agent is the saver. Since $a > \underline{a}$, this necessarily implies that $\underline{\lambda}_4 > 0$, which again implies $\underline{k} = 0$. Therefore, we must have $\lambda_3 > 0$ for the entrepreneur, i.e., the collateral constraint binds for the entrepreneur. This finishes the proof of Lemma A.6. $\square$

Proof of Proposition A.8. Consider the first order conditions (A.13) – (A.17). By Assumption A.5, we have $\lambda_3 > 0$ for the entrepreneur, and $\psi_k \in [0, 1)$. It follows that $\lambda_5 > 0$ and $h = 0$. Moreover, $\underline{\lambda}_5 = 0$ necessarily.

In this case, we obtain by Theorem 3.2 that

$$\frac{1-\alpha}{1-\nu}\frac{C/N}{\alpha} = \frac{r_s}{p_h} = \frac{\underline{\lambda}_2}{\underline{\lambda}_1} = \frac{\underline{a}}{p_k} = \frac{\underline{a}}{A(\psi_k)}\frac{C/N}{\nu}.$$

This implies

$$\underline{a} + (a - \underline{a})\psi_k = A(\psi_k) = \underline{a}\frac{\alpha}{1-\alpha}\frac{1-\nu}{\nu}$$

which can be rewritten as

$$\left[A(\psi_k) + \underline{a}\frac{\alpha}{1-\alpha}\right]\nu = \underline{a}\frac{\alpha}{1-\alpha}.$$

We obtain

$$\frac{\alpha}{\nu} = \frac{1-\alpha}{\underline{a}}\left[A(\psi_k) + \underline{a}\frac{\alpha}{1-\alpha}\right] = \alpha + (1-\alpha)\frac{A(\psi_k)}{\underline{a}}. \tag{A.23}$$

$\square$

Proof of Theorem A.9. By the proof of Theorem A.1, there must be an agent for whom $\lambda_3 = \lambda_5 = 0$. It is clear that this agent must be the (representative) saver. Therefore, by equations (A.15) and housing (A.16), for the saver we have

$$\frac{r_s}{p_h} = \frac{\underline{a}\gamma\underline{k}^{\gamma-1}}{p_k}.$$

We may rewrite this as

$$\begin{aligned} g(\psi_k) = \frac{p_h}{p_k} = \frac{r_s}{\underline{a}\gamma}\underline{k}^{1-\gamma} &= \frac{A(\psi_k)}{\underline{a}\gamma}\frac{K}{H}\frac{1-\alpha}{\alpha}[1-\psi_k]^{1-\gamma}K^{1-\gamma} \\ &= \frac{ak^\gamma + \underline{a}\underline{k}^\gamma}{\underline{a}\gamma K}\frac{K}{H}\frac{1-\alpha}{\alpha}[1-\psi_k]^{1-\gamma}K^{1-\gamma} \\ &= \left[a\psi_k^\gamma + \underline{a}\,(1-\psi_k)^\gamma\right][1-\psi_k]^{1-\gamma}\frac{K}{H}\frac{1-\alpha}{\underline{a}\gamma\alpha} \\ &= \frac{1}{\gamma}\left[1-\psi_k + \frac{a}{\underline{a}}\psi_k^\gamma(1-\psi_k)^{1-\gamma}\right]\frac{K}{H}\frac{1-\alpha}{\alpha}. \end{aligned}$$

Let us denote

$$\widetilde{g}(\psi_k) \equiv 1 - \psi_k + \frac{a}{\underline{a}}\psi_k^{\gamma}(1-\psi_k)^{1-\gamma}.$$

Now

$$\widetilde{g}'(\psi_k) = -1 + \frac{a}{\underline{a}}\gamma\left(\frac{1-\psi_k}{\psi_k}\right)^{1-\gamma} - \frac{a}{\underline{a}}(1-\gamma)\left(\frac{\psi_k}{1-\psi_k}\right)^{\gamma}.$$

For any $\gamma < 1$, we have $\widetilde{g}'(\psi_k) \to +\infty$ as $\psi_k \to 0$. Finally, for $\gamma = 1$ we have

$$\widetilde{g}'(0) = \frac{a - \underline{a}}{\underline{a}} > 0,$$

since $a > \underline{a}$. Since $g'(\psi_k)$ is proportional to $\widetilde{g}'(\psi_k)$, this finishes the proof. □

Proof of Theorem A.10. We consider an entrepreneur with an initial portfolio consisting of only capital and bond holdings. Assuming that the credit constraint is characterized by ϕ_{kko}, the net worth of a credit-constrained entrepreneur is then given by

$$n_o = p_{ko}k_o + b_o = p_{ko}k_o - (1-\phi_{kko})p_{ko}k_o = \phi_{kko}p_{ko}k_o. \tag{A.24}$$

Since $k_o = \psi_{ko}K$ and the price of capital is $p_{ko} = \gamma\underline{a}\,\underline{k}_o^{\gamma-1} = \gamma\underline{a}(1-\psi_{ko})^{\gamma-1}K^{\gamma-1}$, we may rewrite this as

$$n_o = \phi_{kko}\gamma\underline{a}(1-\psi_{ko})^{\gamma-1}\psi_{ko}K^{\gamma}.$$

Consider now an unanticipated credit relaxation, reducing the value of $\phi_k = \phi_{kko}$ to some lower value $\phi_{kk1} < \phi_{kko}$. In this new equilibrium, the capital allocation is now characterized by some endogenous ψ_{k1} and the entrepreneur's wealth is now

$$\begin{aligned}
n_1 &= p_{k1}k_o + b_o \\
&= \gamma\underline{a}(1-\psi_{k1})^{\gamma-1}\psi_{ko}K^{\gamma} - (1-\phi_{kko})\gamma\underline{a}(1-\psi_{ko})^{\gamma-1}\psi_{ko}K^{\gamma} \\
&= \phi_{kko}\gamma\underline{a}(1-\psi_{k1})^{\gamma-1}\psi_{ko}K^{\gamma} + (1-\phi_{kko})\gamma\underline{a}\psi_{ko}K^{\gamma}\left[\frac{1}{(1-\psi_{k1})^{1-\gamma}} - \frac{1}{(1-\psi_{ko})^{1-\gamma}}\right].
\end{aligned}$$

On the other hand, assuming that the entrepreneur is still credit-constrained after trading, also (A.24) has to hold with the new parameters:

$$n_1 = \phi_{kk1}p_{k1}k_1,$$

which gives

$$n_1 = \phi_{kk1}\gamma\underline{a}(1-\psi_{k1})^{\gamma-1}\psi_{k1}K^\gamma.$$

Dividing the first equation for n_1 above by this formula gives

$$1 = \frac{\phi_{kko}\psi_{ko}}{\phi_{kk1}\psi_{k1}} + \frac{(1-\phi_{kko})\psi_{ko}}{\phi_{kk1}\psi_{k1}}\left[1 - \left(\frac{1-\psi_{k1}}{1-\psi_{ko}}\right)^{1-\gamma}\right].$$

The claim follows by reorganizing the equation.

□

Proof of Theorem A.11. We consider an entrepreneur with an initial portfolio consisting only of housing h_o and bonds b_o. Since the saver also invests in the bond with unit return, it must be the case that $p_h = r_s$. Therefore, we may write the the entrepreneur net worth, before trading, as

$$n_{\text{pre}} = n_{\text{pre}}(\psi_k) = p_h h_o + b_o = A(\psi_k)K\frac{h_o}{H}\frac{1-\alpha}{\alpha} + b_o.$$

On the other hand, after trading, for a credit-constrained entrepreneur we must have

$$n_{\text{post}} = n_{\text{post}}(\psi_k) = \phi_k p_k k = \phi_k\gamma\underline{a}\,\underline{k}^{\gamma-1}k = \phi_k\gamma\underline{a}(1-\psi_k)^{\gamma-1}\psi_k K^\gamma.$$

By setting $n = n_{\text{pre}} = n_{\text{post}}$, we may use these two equations to solve for the equilibrium capital allocation ψ_k after trading. Due to the nonlinearity of this system of equations, in general, this may have multiple solutions. I proceed to argue for a particular example of such a multiplicity, confirming that both solutions correspond to an economically meaningful equilibrium.

Let us first note that

$$n_{\text{post}}(0) = 0 \quad \text{and} \quad n'_{\text{post}}(0) = \phi_k\gamma\underline{a}K^\gamma.$$

Similarly, we have

$$n_{\text{pre}}(0) = \underline{a}K^\gamma\frac{h_o}{H}\frac{1-\alpha}{\alpha} + b_o \quad \text{and} \quad n'_{\text{pre}}(0) = +\infty.$$

Therefore, for any $\varepsilon > 0$ small enough, by choosing

$$b_o = -\underline{a}K^\gamma\frac{h_o}{H}\frac{1-\alpha}{\alpha} - \varepsilon, \tag{A.25}$$

we can guarantee one solution ψ_{k1} arbitrarily close to 0, with $n_{\text{pre}}(0) < n_{\text{post}}(0) = 0$ and $n_{\text{pre}}(\psi_k) > n_{\text{post}}(\psi_k)$ for every $\psi_k \in (\psi_{k1}, \psi_{k1} + \delta)$ and for some $\delta > 0$ chosen small enough.

To guarantee the existence of a second solution, by the continuity of n_{pre} and n_{post}, we only need to establish that $n_{\text{pre}}(\psi_k^*) < n_{\text{post}}(\psi_k^*)$, where ψ^* is defined via

$$\frac{a}{\underline{a}} = \left(\frac{\psi_k^*}{1 - \psi_k^*}\right)^{1-\gamma}. \tag{A.26}$$

Indeed, if the entrepreneur is not credit-constrained, then the portfolio problems of both agents dictate that

$$\frac{\gamma a k^{\gamma-1}}{p_k} = 1 = \frac{\gamma \underline{a}\, \underline{k}^{\gamma-1}}{p_k},$$

which implies (A.26) for $\psi_k = \psi_k^*$. Therefore, if in equilibrium $\psi_k < \psi_k^*$, the entrepreneur must be credit-constrained. This implies that, if $n_{\text{pre}}(\psi^*) < n_{\text{post}}(\psi^*)$ and $n_{\text{pre}}(\psi_k) > n_{\text{post}}(\psi_k)$ for some $\psi_k > \psi_{k1}$, there must exist another root ψ_{k2} for $n = n_{\text{pre}} = n_{\text{post}}$ on (ψ_{k1}, ψ_k^*); since by the definition of ψ_k^*, the entrepreneur is credit-constrained with the capital allocation characterized by ψ_{k2}, this produces another equilibrium of the model.

Therefore, it remains to establish a sufficient condition for $n_{\text{pre}}(\psi_k^*) < n_{\text{post}}(\psi_k^*)$. However, since $n_{\text{post}}(\psi_k^*)$ does not depend on h_o and every term of $n_{\text{pre}}(\psi_k^*)$ does, we can always guarantee this by choosing h_o small enough.

To recap the argument, we first choose ψ_k^* to satisfy (A.26). We then choose h_o small enough so that

$$A(\psi_k^*) K \frac{h_o}{H} \frac{1-\alpha}{\alpha} - \underline{a} K^\gamma \frac{h_o}{H} \frac{1-\alpha}{\alpha} < \phi_k \gamma \underline{a} (1 - \psi_k^*)^{\gamma-1} \psi_k^* K^\gamma.$$

For b_o chosen as in (A.25), this guarantees $n_{\text{pre}}(\psi_k^*) < n_{\text{post}}(\psi_k^*)$ and $n_{\text{pre}}(0) < n_{\text{post}}(0)$, for every $\varepsilon > 0$. Finally, we choose $\varepsilon > 0$ small enough to guarantee that there exists $\psi_{k1} \in (0, \psi_k^*)$ such that $n_{\text{pre}}(\psi_k) > n_{\text{post}}(\psi_k)$ for every $\psi_k \in (\psi_{k1}, \psi_{k1} + \delta)$, for some $\delta > 0$ chosen small enough. This guarantees the existence of two equilibria, in both of which the entrepreneur is credit-constrained. $\square$

B Empirical Appendix

B.1 Data

My empirical analyses combine two main data sets: (i) Community Reinvestment Act business loan data and (ii) County Business Patterns data on business activity. In addition, I use county-level Zillow house price indices to measure house price growth rates during 2000 – 2022.

B.1.1 Community Reinvestment Act business loan data

Community Reinvestment Act was enacted in 1977 to improve credit access for businesses in low- and moderate-income (LMI) neighborhoods. The law aimed to remove "redlining" practices that discriminated against loan applications from LMI neighborhoods. According to the law, banking institutions are regularly evaluated on how well they meet their communities' requirements and are given a CRA rating (Outstanding, Satisfactory, Needs to Improve, Substantial Noncompliance). These ratings are used by federal banking agencies, for example, when approving new branches or bank mergers.[33]

The Act operates mainly at the level of U.S. census tracts. The median household income in each tract is compared to the median in the corresponding Metropolitan Statistical Area (MSA) or Metropolitan Division (MD). Tracts with a median household income below 50% of the MSA/MD median are considered low-income, and tracts with household income between 50% and 80% of the MSA/MD median are considered moderate-income tracts.

Each banking institution is assigned a CRA Assessment Area to evaluate its performance based on its presence in the neighborhood (e.g., number and location of branches). The Act provides five different sets of evaluation criteria, e.g., depending on the bank's size, describing the requirements for its presence and operations in the LMI neighborhoods within its assessment area. Based on the value of their assets, banks may be considered Small Banks or Intermediate Small Banks (ISB), both of which have different criteria from larger institutions.[34]

[33]The relevant federal banking agencies are the Board of Governors of the Federal Reserve System, the Office of the Comptroller of the Currency (OCC), and the Federal Deposit Insurance Corporation.

[34]The Small Bank asset threshold was set at $250 million in 1995, increased to one billion in 2005, and has then been adjusted annually, and is currently (as of January 1, 2024) set at $1.564 billion. The ISB threshold was initially set at $250 million in 2005, has been adjusted annually since, and is now (as of January 1, 2024) set at $391 million.

As part of the evaluation process, each institution not considered a Small Bank or an ISB must disclose its lending activity annually. These disclosure data are shared publicly at a bank-county level each year since 1996 by the Federal Financial Institutions Examination Council (FFIEC). The data includes the number and total value of small business loans originated by each bank by loan size at origination ($\leq$ \$100,000, between \$100,000 and \$250,000, and between \$250,000 and \$1,000,000), as well as aggregate counts and values for all business loans originated for firms with annual revenue less than \$1,000,000. These bank-county-level business loan data constitute the primary data for my analyses.

B.1.2 County Business Patterns

I merge the CRA loan data at the county-year level with County Business Patterns (CBP) data on employment and establishment counts published by the U.S. Census Bureau. The CBP data includes employment by county, industry and establishment size (measured in number of employees) between 1987 and 2022 as of March of the reported year. In all the analyses, I use the 2-digit National American Industry Classification System (NAICS) level and exclude the residential construction (NAICS 1997 category 2332; NAICS 2002- category 2361) and real estate (NAICS category 53) sectors to ensure that the specifics of the housing boom-bust cycles do not drive my results.

In some small counties, employment data for all industries is not disclosed due to privacy reasons, and only a size class is provided. In such cases, I use the midpoint of the category, except for "over 100,000 employees," where 200,000 employees are used as a proxy for the employment count. The data also includes the full county-level employment and establishment counts. However, in my regression specifications, I only use industry-level data, which allows for removing the residential construction and real estate sectors.

Table 4: Price-to-rent ratio and residential investment

	Dependent variable:							
	Price-to-rent ratio				Price-to-rent ratio with forced intercept 1.0			
	Contemporaneous	*12 mos lag*	*15 mos lag*	*18 mos lag*	*Contemporaneous*	*12 mos lag*	*15 mos lag*	*18 mos lag*
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Investment rate	2,100.2***	1,716.2***	1,664.1***	1,488.9***	5,628.1***	5,890.0***	5,971.3***	6,070.1***
	(197.6)	(197.1)	(160.7)	(189.7)	(72.2)	(113.1)	(123.2)	(147.0)
Constant	7.8***	8.8***	8.9***	9.3***	1.0	1.0	1.0	1.0
	(0.4)	(0.4)	(0.3)	(0.4)				
Observations	26	27	27	27	26	27	27	27
R^2	0.792	0.849	0.870	0.841	0.996	0.991	0.989	0.985
Adjusted R^2	0.783	0.843	0.864	0.835	0.996	0.990	0.989	0.984
Note:								*p<0.005; **p<0.001; ***p<5e-04

Regression of quarterly price-to-rent ratio on investment rate. Price-to-rent ratio is constructed from Zillow U.S. rental and price data. The price-to-rent ratio data is available since the beginning of 2014, and these time-series regressions are for price-to-rent ratios measured during 2014 – 2020. Investment rate is the growth rate of housing units in the U.S.A measure by the U.S. Census Bureau. Standard errors, shown in parentheses, are Newey-West (1987) heteroskedastic robust.

Table 5: Small business loan originations and house prices

The table reports the results for an annual county-level regression of growth in small business loan originations ($\leq$ \$250,000 at origination) on house price growth rates during 2000 – 2022. The loan origination counts are from Community Reinvestment Act data and house prices from the Zillow house price indices at the county level. House price growth is instrumented by interacting the Saiz (2010) housing supply elasticity measure with the 30-year mortgage rate. Standard errors are clustered at the state level, and the regressions are weighted by county-level (log) population in 2010. Controls include the county-level share of the white population, homeownership rate, share of the population in the labor force, unemployment rate, and share of the population with a college degree, all measured in 2010. The loan growth data is restricted only to include loan growth rates between the 10th and 90th percentiles to remove outliers and infinite growth rates (associated with zero originations).

	Dependent variable:					
	House price growth	Growth in loan originations				
	First stage	*Reduced form*	*OLS*		*Instrumental variable (2SLS)*	
	(1)	(2)	(3)	(4)	(5)	(6)
Housing supply elasticity	-0.0005***	-0.0010***				
× 30-year mortgage rate	(0.0001)	(0.0002)				
House price growth			0.119***	0.116***	1.91***	1.74***
			(0.012)	(0.012)	(0.412)	(0.404)
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Controls	No	No	No	Yes	No	Yes
Observations	13,431	13,305	38,454	38,454	13,431	13,431
R^2	0.564	0.572	0.359	0.360		

Clustered (state) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Throughout the paper, I restrict both the CRA loan growth rates and the CBP employment and establishment growth rates between the 10th and 90th percentiles to remove infinite values and other outliers.[35]

B.2 Stylized facts

Adelino et al. (2015) argue that increases in real estate prices affect employment growth by facilitating collateralized external financing and creating or expanding small business activity via the collateral channel. They consider the differential effect of house prices on the expansion of employment between establishments of different sizes: assuming small business financing is more likely to rely on real estate collateral, house prices should have a differential impact on small business expansion compared to large corporations. In order to reason for a causal effect, they instrument house prices by the housing supply elasticity, measured by land topography differences by Saiz (2010). In addition to the effect of house prices on employment, Adelino et al. (2015) document the intermediary step, by showing the effect of house prices on home equity refinancing and on home equity line of credit loans. For business loans they consider commercial and industrial loans reported in bank Call Reports. However, for these they do not find a statistically significant effect. While arguing for the same collateral effect of Adelino et al. (2015), I augment their analysis by considering small business loan originations in the CRA business loan data. I show that unlike in bank Call Reports, for the CRA business loan originations, we find similar effects than Adelino et al. (2015) find for the home equity loans.

B.2.1 House prices and small business loans

In Panel A of Figure 2, I plot the Cattaneo et al. (2024) binscatter between growth rates in small business loan originations ($\leq$ \$250,000 at origination) and house prices. On the left, I depict this across U.S. counties using county-level panels from 2000 to 2022. On the right, I leverage the bank-level data and depict the binscatter using the bank-county level panel data from 2000 to 2022. The figure demonstrates a strong correlation between these variables, which is consistent with the findings of Adelino et al. (2015). I validate this relationship using two different regression specifications. First, I consider a county-year-level regression reported in Table 5. I instrument the

[35] Especially in the bank-level data, there are many bank-county pairs with zero observations in some particular years. These lead to infinite growth rates, which are removed by this sample restriction, which is applied throughout the paper.

house price growth rates with the IV strategy of Chaney et al. (2012), interacting the MSA-level Saiz (2010) land topography housing supply elasticity measure with the 30-year mortgage rates. The first stage of the IV specification is given by:[36]

$$\text{House price growth}_{it} = \alpha_j + \lambda_t + \rho\, \text{Elasticity}_k \times \text{Mortgage rate}_t + \gamma' X_i + \varepsilon_{it}, \quad \text{(B.1)}$$

where α_j are state fixed effects, λ_t are year fixed effects, housing supply elasticity is measured at the MSA-level (k), and X_i is a vector of controls, including the county-level share of the white population, homeownership rate, share of the population in the labor force, unemployment rate, and share of the population with a college degree, all measured in 2010. The regression is weighted by each county's 2010 (log) population, and standard errors are clustered at the state level.

I specify the second state of the IV regression as follows:

$$\text{Loan origination growth}_{it} = \theta_j + \delta_t + \beta\, \text{House price growth}_{it} + \nu' X_i + \varepsilon_{it}, \quad \text{(B.2)}$$

where θ_j are state and δ_t year fixed effects. In both regressions, I restrict the sample between the 10th and 90th percentiles of loan growth rates to remove infinite (or otherwise large) growth rates stemming from small counties with zero loan originations in any particular year.

The OLS regressions re-establish the strong correlation in Panel A of Figure 2, whereas the IV setup suggests a large elasticity of 1.91. Notably, this estimated elasticity has the same order of magnitude as reported by Adelino et al. (2015) for home equity refinancing (1.60) and for cash-out HELOC loans (1.72).

To provide further confidence on the large estimated elasticity from house price growth rates to business loan originations, I augment this analysis by considering the CRA bank-level panel data on small business loan originations, with the second stage regression specified as

$$\text{Loan origination growth}_{it} = \eta_i + \theta_j + \delta_t + \beta\, \text{House price growth}_{lt} + \nu' X_l + \varepsilon_{it}, \quad \text{(B.3)}$$

where η_i are bank, θ_j state and δ_t year fixed effects, house price growth rates are measured at the county level (l), and X_l is the vector of county-level controls, as before. This specification estimates the within-bank elasticity, which excludes the effects on the extensive margin when new banks enter and exit the market during the business cycle. Therefore, we should expect the estimated elasticity to be lower than

[36] All growth rates are measured as log differences.

Table 6: Small business loan originations and house prices at the bank level

The table reports the results for an annual bank-county-level regression of growth in small business loan originations ($\leq$ \$250,000 at origination) on house price growth rates during 2000 – 2022. The loan origination counts are from Community Reinvestment Act data and house prices from the Zillow house price indices at the county level. House price growth is instrumented by interacting the Saiz (2010) housing supply elasticity measure with the 30-year mortgage rate. Standard errors are clustered at the bank level, and the regressions are weighted by bank-level (log) small business loan origination counts. Controls include the county-level share of the white population, homeownership rate, share of the population in the labor force, unemployment rate, and share of the population with a college degree, all measured in 2010. The loan growth data is restricted only to include loan growth rates between the 10th and 90th percentiles to remove outliers and infinite growth rates (associated with zero originations).

	Dependent Variable: Growth in loan originations			
	OLS		*IV (2SLS)*	
	(1)	(2)	(3)	(4)
House price growth	0.082***	0.082***	1.00**	1.03***
	(0.023)	(0.023)	(0.394)	(0.349)
Bank FE	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Controls	No	Yes	No	Yes
Observations	1,002,521	1,002,521	491,509	491,509
R^2	0.088	0.088		

Clustered (bank) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

in the county-level analysis, as (B.3) reflects only the intensive margin effects. On the other hand, including bank fixed effects in the regression should provide further confidence in the strong connection between collateral valuations and business loan originations.[37]

I report the results of this regression in Table 6; column (3) documents the elasticity of a 1.1 (log) point increase in the growth of within-bank loan originations for a one (log) point increase in the house price growth rate. Column (4) of the table demonstrates the robustness of the result for including a set of county-level controls. The drop in the elasticity from 1.9 to 1.1 between the county- and bank-level analyses suggests a substantial role for the extensive margin as new banks enter and exit the small business loan ($\leq$ \$250,000 at origination) market during the business cycle.

B.2.2 Small business loans and business activity

The second key component of the collateral channel concerns the effect of business loans on business activity, which I measure by employment and number of establishments at the county-year level. Panel B of Figure 2 demonstrates a strong correlation between business loan origination and both measures of business activity.

In section B.2.1, I argued that house prices have a robust aggregate effect on business loan originations, consistent with existing micro-level evidence on collateral valuations' role in entrepreneurial financing and investments (Schmalz et al., 2017; Chaney et al., 2012; Bahaj et al., 2020). In this case, the validity of the

[37] To implement this regression, I restrict the data by only including observations, where the loan growth rate is between the 10th and 90th percentiles.

Table 7: Business loan originations and employment

The table reports the results for an annual county-level regression of growth in employment on growth in small business loan originations ($\leq$ \$250,000 at origination) during 2000 – 2022. The loan origination counts are from Community Reinvestment Act data. Employment information collected from County Business Patterns excludes construction and real estate sectors. I use the Zillow house price indices at the county level to control house prices. Growth in loan originations is instrumented by interacting the Saiz (2010) housing supply elasticity measure with the 30-year mortgage rate. Standard errors are clustered at the state level. The regressions are weighted by county-level (log) population in 2010. The loan and employment growth rates are restricted between the 10th and 90th percentiles.

	Dependent variable: Employment growth						
	Reduced form	*OLS*			*Instrumental variable (2SLS)*		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Housing supply elasticity	-0.0005***						
$\times$ 30-year mortgage rate	(9.02×10^{-5})						
Growth in loan originations		0.016***	0.015***	0.013***	0.480***	0.472***	0.379***
		(0.003)	(0.003)	(0.003)	(0.099)	(0.107)	(0.136)
House price growth			0.073***	0.069***		0.016	0.028
			(0.008)	(0.008)		(0.022)	(0.023)
State FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	No	No	No	Yes	No	No	Yes
Observations	9,244	23,673	23,673	23,673	9,244	9,244	9,244
R^2	0.156	0.077	0.085	0.101			

Clustered (state) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 8: Business loan originations and number of establishments

The table reports the results for an annual county-level regression of growth in number of establishments on growth in small business loan originations ($\leq$ \$250,000 at origination) during 2000 – 2022. The loan origination counts are from Community Reinvestment Act data. Information on the number of establishments collected from County Business Patterns excludes construction and real estate sectors. I use the Zillow house price indices at the county level for house prices. Growth in loan originations is instrumented by interacting the Saiz (2010) housing supply elasticity measure with the 30-year mortgage rate. Standard errors are clustered at the state level. The regressions are weighted by county-level (log) population in 2010. The loan and establishment growth rates are restricted between the 10th and 90th percentiles.

	Dependent variable: Growth in number of establishments						
	Reduced form	*OLS*			*Instrumental variable (2SLS)*		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Housing supply elasticity	-0.0003***						
× 30-year mortgage rate	(7.84×10^{-5})						
Growth in loan originations		0.014***	0.013***	0.012***	0.375***	0.380***	0.276**
		(0.001)	(0.001)	(0.001)	(0.093)	(0.103)	(0.123)
House price growth			0.050***	0.048***		-0.008	0.009
			(0.004)	(0.004)		(0.023)	(0.023)
State FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	No	No	No	Yes	No	No	Yes
Observations	12,903	35,385	35,385	35,385	12,903	12,903	12,903
R^2	0.280	0.166	0.178	0.207			

Clustered (state) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

housing supply elasticity instrument for house price growth rates - argued, e.g. by Mian and Sufi (2011) - should suggest that the supply elasticity measure can also instrument business loan originations. Thus, I consider the correlations of panel B of Figure 2 in regression setups using the housing supply elasticity as an instrument. In my analyses, I exclude real estate and residential construction sectors to ensure that the specifics of the housing boom-bust cycles during my study period do not drive my results and to alleviate some of the concerns raised by Kerr et al. (2022).

Loan originations and employment. I regress the annual growth rate in employment on the growth in small business loan originations ($\leq$ \$250,000 at origination) across U.S. counties during 2000 – 2022:

$$\text{Employment Growth}_{it} = \theta_j + \delta_t + \beta \, \text{Loan growth}_{it} + \nu' X_{it} + \varepsilon_{it},$$

where δ_j are state and θ_t year fixed effects, and the vector of controls X_{it} is as in section B.2.1, augmented with annual house price growth rates in some regression specifications reported in Table 7.

The first stage regression follows the structure of (B.1) and is given by[38]

$$\text{Loan growth}_{it} = \alpha_j + \lambda_t + \rho \, \text{Elasticity}_k \times \text{Mortgage rate}_t + \gamma' X_i + \varepsilon_{it}. \tag{B.4}$$

The results for this first stage are reported in column (2) of Table 5, excluding controls. We observe a statistically significant negative coefficient, documenting a negative association between housing supply elasticity and loan growth rates. This is consistent with the expectation of inelastic areas enjoying higher house prices and (collateralized) loan growth rates.

This IV regression of employment growth on loan growth rates documents an elasticity between 0.379 and 0.480, depending on the exact regression specification. In the OLS regressions of columns (3) and (4) of Table 7, house price growth rates are highly statistically significant when introduced as controls. Nevertheless, the coefficient on loan originations remains stable. Most notably, however, when instrumenting the loan origination growth rates with housing supply elasticity, the residual correlation of house prices for employment growth drops to zero. This provides further evidence for the results of Adelino et al. (2015), arguing that the collateral channel provides an essential source of external financing for small businesses, with aggregate implications for employment growth rates.

[38] I restrict the loan and employment data between the 10th and 90th percentiles.

Note that the instrumented elasticity in columns (5) to (7) are substantially higher than the OLS estimates in columns (2) to (4) of Table 7. Similarly to the analyses in section 6, this is likely due to two reasons. First, for privacy reasons, the County Business Patterns data only provides “employment flags” showing the size class of the county-industry pair for such pairs, where there are not enough firms to maintain the anonymity of the data. In my analysis, which uses industry-level data to exclude the real estate and residential construction sectors, I encode these employment flags using the midpoint of the specified employment class. This produces measurement error generating attenuation bias.

Second, the OLS estimate likely reflects the fact that the Saiz IV captures the local average treatment effect (LATE) among the complier counties that respond to the housing supply elasticity “shock” (i.e. to the exogenous variation in supply elasticity). For example, through the lens of my theory, the house price increase is due to improved capital allocation, when entrepreneurs use the new collateral capacity associated with the credit relaxation (Duca, Muellbauer, and Murphy, 2011; Levitin and Wachter, 2011) to purchase business capital. However, there may be natural cross-regional differences in business dynamism; high-growth areas, e.g., in San Francisco or New York, may have more entrepreneurial activity and, therefore, have a differential ability to benefit from such a credit shock compared to other low-supply elasticity areas such as Fort Lauderdale or Los Angeles. In such a case, the IV specification only captures the local average treatment effect among those complier counties that, in reality, have enough entrepreneurs who can benefit from the credit relaxation. In particular, this implies that we should interpret these estimates with cautiousness, as the large local average treatment effect does not necessarily imply a large average treatment effect.

Loan originations and business creation. In addition to employment, we can consider the effect of loan originations on business creation measured by the annual growth rate in number of establishments across U.S. counties:[39]

$$\text{Growth in \# of establishments}_{it} = \theta_j + \delta_t + \beta\,\text{Loan growth}_{it} + \nu' X_{it} + \varepsilon_{it},$$

where θ_j and δ_t are again state and year fixed effects, respectively, and X_{it} is a vector of controls as in (B.2.1). The first stage for the IV regression is still given by (B.4).

I report the results in Table 8. The elasticity for the growth rate in the number of establishments with respect to the growth in small business loan originations is

[39] Again, I restrict the loan data and the data on establishment growth rates between the 10th and 90th percentiles.

estimated between 0.276 and 0.375, depending on the exact regression specification (with or without controls). In broad terms, this is similar to the one for employment.

Notably, after instrumenting the loan growth rates with the housing supply elasticity, the residual effect of house prices on business creation is zero, again similar to the employment effects reported in Table 7. Finally, I note that the same caveats related to the substantial difference between the OLS and IV estimates, as discussed in relation to the employment effects, also apply here (cf. measurement error and attenuation bias, and LATE estimates).

C Dynamic model

C.1 Notation

$$\psi_{ht} = \frac{h_t}{h_t + \underline{h}_t} \quad \text{and} \quad \psi_{kt} = \frac{k_t}{k_t + \underline{k}_t}$$

and

$$\nu_t = \frac{p_{kt}K}{p_{kt}K + p_{ht}H}$$

Net worth

$$\underline{n}_t = p_{ht}\underline{h}_t + p_{kt}\underline{k}_t + \underline{b}_t \quad \text{and} \quad n_t = p_{ht}h_t + p_{kt}k_t + b_t,$$

Portfolio shares

$$\underline{\theta}_{ht} = \frac{p_{ht}\underline{h}_t}{\underline{n}_t} \quad \text{and} \quad \underline{\theta}_{kt} = \frac{p_{kt}\underline{k}_t}{\underline{n}_t}.$$

Saver wealth share

$$\eta_t = \frac{n_t}{n_t + \underline{n}_t}.$$

C.2 Asset market clearing

$$\underline{\theta}_{ht} = \frac{1 - \psi_{ht}}{1 - \eta_t}(1 - \nu_t)$$

$$\underline{\theta}_{kt} = \frac{1 - \psi_{kt}}{1 - \eta_t}\nu_t$$

$$\theta_{ht} = \frac{\psi_{ht}}{\eta_t}(1 - \nu_t)$$

$$\theta_{kt} = \frac{\psi_{kt}}{\eta_t}\nu_t.$$

LTV constraint:

$$\frac{\psi_{ht}}{\eta_t}(1-\nu_t) + \frac{\phi_k}{\phi_h}\frac{\psi_{kt}}{\eta_t}\nu_t \le \frac{1}{\phi_h}.$$

The boundary of the binding set is characterized by $\eta = \phi_k \nu$.

C.3 Hamilton-Jacobi-Bellman equation

The Hamilton-Jacobi-Bellman equation associated with this problem is given by

$$\begin{aligned} &\rho V(n;\mathbf{S}) \\ &= \max_{c,s,\theta_k,\theta_h} u(c,s) + \nabla_{\mathbf{S}} V \cdot d\mathbf{S} \\ &\quad + \frac{\partial V}{\partial n} n \left[-\frac{c}{n} - \frac{r_s s}{n} + r_f + \theta_k \left(\frac{a}{p_k} + \mu_k^p - r_f \right) + \theta_{ht} \left(\frac{r_s}{p_k} + \mu_h^p - r_f \right) \right] \\ &\text{s.t.} \quad \theta_k + \frac{\theta_h}{\phi_k} \le \frac{1}{\phi_k}, \quad \theta_k \ge 0, \quad \theta_h \ge 0. \end{aligned} \tag{C.1}$$

where $\nabla_{\mathbf{S}} V$ denotes the gradient of V with respect to $\mathbf{S}$.

C.4 Proofs

Denote the multiplier on the collateral constraint by λ^{LTV} and the multipliers on the nonnegativity constraints by λ^k and λ^h, respectively. I suppress the aggregate state variable from the value function and denote $V'(n) := \frac{\partial V}{\partial n}$, The FOCs are

$$\begin{aligned} \frac{a}{p_k} + \mu_k^p &= r^f + \frac{\lambda^{LTV} - \lambda^k}{V'(n)n}, \\ \frac{r_s}{p_h} + \mu_h^p &= r^f + \frac{\lambda^{LTV}/\phi_k - \lambda^h}{V'(n)n}, \\ u_c &= V'(n), \\ u_s &= V'(n) r_s. \end{aligned}$$

It immediately follows that

$$\frac{u_s}{u_c} = r_s.$$

For the utility specification $u(c^\alpha s^{1-\alpha})$ this gives

$$\frac{1-\alpha}{\alpha} c = r_s s \implies r_s = A(\psi_k)\frac{K}{H}\frac{1-\alpha}{\alpha}, \tag{C.2}$$

where I used the goods market clearing $C = A(\psi_k)K$.

Moreover, by the goods market clearing we also have

$$C = A(\psi_k)K \implies \frac{C}{N} = \frac{A(\psi_k)}{p_k}\frac{p_k K}{N},$$

which gives

$$p_k = \frac{A(\psi_k)}{C/N}\nu \implies p_h = \frac{A(\psi_k)}{C/N}\frac{K}{H}(1-\nu). \tag{C.3}$$

This implies that the rental yield is given by

$$\frac{r_s}{p_h} = \frac{1-\alpha}{1-\nu}\frac{1}{\alpha}\left[\frac{c_t}{n_t}\eta_t + \frac{\underline{c}_t}{\underline{n}_t}(1-\eta_t)\right].$$

Now

$$\mu_k^p = \frac{\dot{p}_k}{p_k} = \frac{1}{p_k}\left[\frac{d}{dt}\left[\frac{A(\psi_k)}{C/N}\right]\nu + \frac{A(\psi_k)}{C/N}\dot{\nu}\right]$$

and

$$\mu_h^p = \frac{\dot{p}_k}{p_k} = \frac{1}{p_h}\left[\frac{d}{dt}\left[\frac{A(\psi_k)}{C/N}\right]\frac{K}{H}(1-\nu) - \frac{A(\psi_k)}{C/N}\frac{K}{H}\dot{\nu}\right],$$

which implies

$$\mu_k^p - \mu_h^p = \frac{\dot{\nu}}{\nu} + \frac{\dot{\nu}}{1-\nu} = \frac{\dot{\nu}}{\nu(1-\nu)}$$

at points of differentiability.

The dynamics for ν in Theorem 3.3 follow. Moreover, in the unconstrained region

$$\begin{aligned}\frac{r_s}{p_h} - \frac{a}{p_k} &= \left[\frac{1/\alpha - 1}{1-\nu} - \frac{1}{\nu}\right]\left[\frac{c_t}{n_t}\eta_t + \frac{\underline{c}_t}{\underline{n}_t}(1-\eta_t)\right] \\ &= \frac{\frac{\nu}{\alpha}-1}{\nu(1-\nu)}\left[\frac{c_t}{n_t}\eta_t + \frac{\underline{c}_t}{\underline{n}_t}(1-\eta_t)\right]\end{aligned}$$

and in the constrained region

$$\begin{aligned}\frac{r_s}{p_h} - \frac{\underline{a}}{p_k} &= \left[\frac{1/\alpha - 1}{1-\nu} - \frac{\underline{a}}{A(\psi_k)\nu}\right]\left[\frac{c_t}{n_t}\eta_t + \frac{\underline{c}_t}{\underline{n}_t}(1-\eta_t)\right] \\ &= \left[\frac{1/\alpha - 1}{1-\nu} - \frac{1 - \frac{a-\underline{a}}{A(\psi_k)}\psi_k}{\nu}\right]\left[\frac{c_t}{n_t}\eta_t + \frac{\underline{c}_t}{\underline{n}_t}(1-\eta_t)\right] \\ &= \left[\frac{\nu/\alpha - 1 + \frac{a-\underline{a}}{A(\psi_k)}\psi_k(1-\nu)}{\nu(1-\nu)}\right]\left[\frac{c_t}{n_t}\eta_t + \frac{\underline{c}_t}{\underline{n}_t}(1-\eta_t)\right].\end{aligned}$$

It follows that necessarily $\nu < \alpha$ along the saddle path in the constrained region, if $\nu_t \to \alpha$ as $t \to \infty$.

Moreover,

$$dn_t = -c_t - r_{st}s_t + r_{ft}n_t + \frac{\lambda^{LTV}}{\phi_k}n_t$$

and similarly for $d\underline{n}_t$. It follows that

$$\frac{d(n+\underline{n})}{n+\underline{n}} = \eta\frac{dn}{n} + (1-\eta)\frac{d\underline{n}}{\underline{n}}\,dt = \eta\left[r_{ft} - \frac{c_t}{n_t} - \frac{r_{st}s_t}{n_t} + \frac{\lambda^{LTV}}{\phi_k V'(n)n}\right] + (1-\eta)\left[r_{ft} - \frac{\underline{c}_t}{\underline{n}_t} - \frac{r_{st}\underline{s}_t}{\underline{n}_t}\right].$$

We obtain

$$\begin{aligned}\frac{d\eta}{\eta} = \frac{d\left(\frac{n}{n+\underline{n}}\right)}{\frac{n}{n+\underline{n}}} &= r_{ft} - \frac{c_t}{n_t} - \frac{r_{st}s_t}{n_t} + \frac{\tau}{\phi_k} - \frac{d(n+\underline{n})}{n+\underline{n}} \\ &= (1-\eta)\left[\frac{\underline{c}_t}{\underline{n}_t} - \frac{c_t}{n_t} + \frac{\tau}{\phi_k}\right] + (1-\eta)\left[\frac{r_{st}\underline{s}_t}{\underline{n}_t} - \frac{r_{st}s_t}{n_t}\right],\end{aligned}$$

where $\tau \equiv \frac{\lambda^{LTV}}{V'(n)n}$.

C.5 Log utility

Guess and verify

$$V(n;\boldsymbol{S}) = \frac{1}{\rho}\log\left[\omega(\boldsymbol{S})n\right].$$

which implies

$$V'(n) = \frac{1}{\rho n} \implies c = \alpha\rho n, \quad r_s s = (1-\alpha)\rho n.$$

This implies that

$$\dot{\eta} = \eta(1-\eta)\left[\underline{\rho} - \rho + \frac{\tau}{\phi_k}\right],$$

which fully determines the model dynamics in the unconstrained region $\eta > \phi_k\nu$.

C.5.1 The constrained region: $\eta < \phi_k\nu$.

Now by the collateral constraint

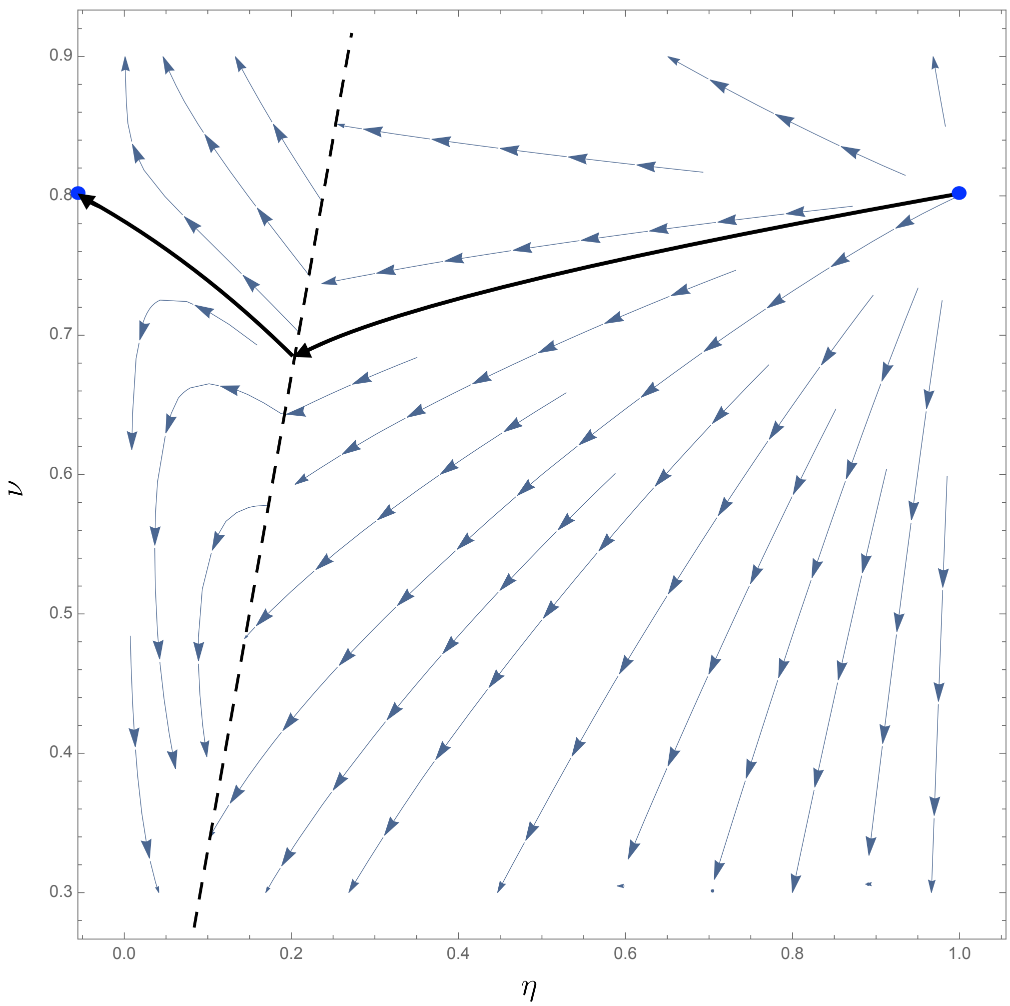
$$\psi_k = \frac{\eta}{\phi_k\nu}$$

and by the FOCs

$$\tau = \frac{a - \underline{a}}{p_k}.$$

These together fully determine the dynamics in the constrained region.

Figure 11: Phase diagram



The picture depicts the (η, ν) phase diagram for the case (a) of Proposition 4.5. The solid black line shows the saddle path, which connects the non-stable steady-state at $(\eta, \nu) = (1, \alpha)$ to the dynamically stable one at $(\eta, \nu) = (0, \alpha)$. The steady-states are shown in blue dots. The dashed line separates the two regions $\{\eta < \phi_k \nu\}$ and $\{\eta \geq \phi_k \nu\}$.

D Modeling details for the Quantitative Analysis of Section 5

By the goods market clearing we have

$$C = A(\psi_k)K - i_k K = \frac{A(\psi_k)}{p_k} p_k K - \frac{p_k - 1}{\kappa_k} K = \frac{A(\psi_k)}{p_k} p_k K + \frac{p_k K}{\kappa_k p_k} - \frac{p_k K}{\kappa_k}.$$

This implies which gives

$$p_k = \frac{1 + \kappa_k A(\psi_k)}{\nu + \kappa_k C/N} \nu \tag{D.1}$$

as well as

$$p_h = \frac{1 + \kappa_k A(\psi_k)}{\nu + \kappa_k C/N} \frac{K}{H} (1 - \nu) \quad \text{and} \quad \frac{p_h}{r_s} = \frac{1 + \kappa_h}{1 - \nu + \kappa_h \frac{r_s S}{N}} (1 - \nu).$$

The investment rates are given by

$$i_k = \frac{A(\psi_k)\nu - C/N}{\nu + \kappa_k C/N} \quad \text{and} \quad i_h = \frac{1 - \nu - \frac{r_s S}{N}}{1 - \nu + \kappa_h \frac{r_s S}{N}}.$$

Denote the multiplier on the collateral constraint by λ^{LTV} and the multipliers on the nonnegativity constraints by λ^k and λ^h, respectively. I suppress the state variable from the value function and denote $V'(n) := \frac{\partial V}{\partial n}$, The FOCs are

$$\begin{aligned}
\frac{a - i_k}{p_k} + g_k + \mu_k^p &= r^f + \frac{\lambda^{LTV} - \lambda^k}{V'(n) n}, \\
\frac{1 - i_h}{p_h / r_s} + g_h + \mu_h^p &= r^f + \frac{\lambda^{LTV}/\phi_k - \lambda^h}{V'(n) n}, \\
u_c &= V'(n), \\
u_s &= V'(n) r_s,
\end{aligned}$$

where

$$g_k := \phi_k(i_k) - \delta_k \quad \text{and} \quad g_h = \phi_h(i_h) - \delta_h.$$

It immediately follows that

$$\frac{u_s}{u_c} = r_s.$$

For the utility specification $u(c^\alpha s^{1-\alpha})$ this gives

$$\frac{1-\alpha}{\alpha} c = r_s s. \tag{D.2}$$

By (D.2) we also have

$$\begin{aligned}
r_s = \frac{A(\psi_k) - i_k}{1 - i_h} \frac{K}{H} \frac{1-\alpha}{\alpha} &= \frac{\frac{\left[1+\kappa_k A(\psi_k)\right] C/N}{\nu + \kappa_k C/N}}{\frac{[1+\kappa_h]\frac{r_s S}{N}}{1-\nu+\kappa_h \frac{r_s S}{N}}} \frac{K}{H} \frac{1-\alpha}{\alpha} \\
&= \frac{\left[1+\kappa_k A(\psi_k)\right] C/N}{[1+\kappa_h]\frac{r_s S}{N}} \frac{1-\nu+\kappa_h \frac{r_s S}{N}}{\nu + \kappa_k C/N} \frac{K}{H} \frac{1-\alpha}{\alpha} \\
&= \frac{1+\kappa_k A(\psi_k)}{1+\kappa_h} \frac{1-\nu+\kappa_h \frac{r_s S}{N}}{\nu + \kappa_k C/N} \frac{K}{H},
\end{aligned}$$